

Research Papers

Correlation-aware fusion model for robust SOH estimation in lithium-ion batteries

Yubin Gong^a, Xinshu Lu^a, Zixiang Su^a, Zhen Huang^c, Yong Quan^c, Zhiqiang Zhang^c,
Tiancheng Ouyang^{a,b,*}

^a School of Mechanical Engineering, Guangxi Colleges and Universities Engineering Research Center of Advanced Powertrain for New Energy Vehicles, Guangxi University, Nanning, PR China

^b School of Mechanical Engineering, Southeast University, Nanjing, PR China

^c Dongfeng Liuzhou Automobile Co., Ltd, Liuzhou, PR China

ARTICLE INFO

Keywords:

Lithium-ion batteries
Kolmogorov-Arnold networks-transformer
Health indicator
Pearson correlation coefficient
State of health

ABSTRACT

The generalizability of data-driven models is crucial for lithium-ion battery state of health (SOH) estimation, relying on degradation information embedded in selected health indicators (HIs). Most studies focus on HIs with high Pearson correlation coefficient (PCC) to improve prediction accuracy, often overlooking potential degradation information in low-PCC HIs that supports generalization. To address this limitation, this study proposes a dual-path correlation-aware fusion model integrating a Kolmogorov-Arnold Network-transformer (KAT), convolutional neural networks (CNN), and long short-term memory (LSTM) networks, designed to extract hidden degradation features from low-PCC HIs while enhancing generalization across datasets. Comprehensive experiments on three battery datasets demonstrate the method's robustness and predictive precision, with maximum mean absolute error (MAE) and root mean square error (RMSE) of only 0.89 % and 0.97 %. Furthermore, cross-dataset evaluation confirms that the proposed model maintains accurate and robust SOH estimation under varying temperatures, C-rates, and measurement platforms, highlighting its strong generalization capability.

1. Introduction

Lithium-ion batteries, as a core element of electric vehicles (EVs), have become a rapidly expanding focus in both chemical and physical research domains [1]. However, lithium battery life degradation mechanisms are complex, and their prediction and health management (PHM) are critical, and serious safety accidents can occur if lithium batteries are not managed properly [2]. Accurate estimation of the state of health (SOH) of lithium-ion batteries constitutes a key task in the battery management system (BMS) of electric vehicles [37]. Ensuring reliable estimation performance across various battery chemistries and operating conditions, namely the generalization capability of SOH estimation, remains a significant and ongoing challenge. Accurate estimation of SOH also can help the BMS to correctly determine the aging state of the battery, improve fault prediction performance and ensure the safe operation of EVs [3,4].

Currently, SOH estimation is mainly categorized into model-based [5] and data-driven approaches [6]. The model-based approach

ensures accurate estimation of battery SOH by explicitly modeling the internal states of lithium batteries and incorporating some filtering algorithms [7,8]. A method proposed by Xiong et al. [9] estimates SOH for lithium-ion batteries in all-climate EVs based on electrochemical models (EMs), where partial differential equations are employed to simulate aging behavior, and degradation trends are obtained through parameter extraction and numerical computations. However, in EM, the partial differential equations are complex to solve, the parameters are challenging to identify, and a substantial amount of experimental data is required for validation. Accordingly, SOH estimation often relies on the equivalent circuit model (ECM) due to its modeling efficiency and ease of implementation. Chen et al. [10] developed an SOH estimation method by coupling a fractional-order ECM with particle swarm optimization to identify key internal battery parameters. Considering the effect of discharging rate on SOH, Xu et al. [11] proposed an estimation framework that accounts for the discharging rate by integrating a fractional-order ECM with a dual extended Kalman filter, significantly enhancing estimation accuracy. While ECM substantially reduces the complexity of EM, such simplification inevitably compromises model

* Corresponding author at: School of Mechanical Engineering, Guangxi Colleges and Universities Engineering Research Center of Advanced Powertrain for New Energy Vehicles, Guangxi University, Nanning, PR China.

E-mail address: ouyangtiancheng@gxu.edu.cn (T. Ouyang).

<https://doi.org/10.1016/j.est.2025.120018>

Received 27 August 2025; Received in revised form 26 October 2025; Accepted 19 December 2025

Available online 8 January 2026

2352-152X/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Acronyms

HI	Health indicator
EV	Electric vehicle
SOH	State of health
PCC	Pearson correlation coefficient
KAN	Kolmogorov-Arnold networks
KAT	Kolmogorov-Arnold networks-transformer
CNN	Convolutional neural networks
LSTM	Long and short-term memory
MLP	Multi-layer perceptron
GRU	Gated recurrent unit
ECM	Equivalent circuit model
EM	Electrochemical model
MAE	Mean absolute error
RMSE	Root mean square error
BMS	Battery management system

generalizability and exacerbates the propagation of errors in SOH estimation [12]. In summary, model-based approaches ensure physical interpretability but are difficult to scale to diverse operating conditions due to complex parameterization and strong dataset dependence.

The popularity of data-driven methods in SOH estimation stems from their capacity to learn degradation behavior directly from input features, bypassing the need for detailed modeling of the battery's internal mechanisms, including linear regression [13], support vector machines [14], and deep learning [15,16]. With respect to generalization, data-driven methods offer a distinct advantage over model-based approaches by eliminating the need for repeated parameter adjustments related to battery type, state of charge (SOC) windows, and other factors [38]. Data-driven methods are capable of directly extracting shared features from cross-condition datasets, thereby substantially improving model generalizability. Deep learning stands as one of the most widely adopted methodologies, including convolutional neural networks (CNN) [17], gated recurrent unit (GRU) [18], transformer [19] and so on. Li et al. [20] developed a hybrid model incorporating fiber bragg grating sensors and GRU, where a CNN extract features from electrical and thermal signals, and GRU captures the temporal relationships for SOH prediction. Although this method effectively integrates multimodal information, it depends on additional sensing hardware, limiting its scalability for large-scale applications. Long short-term memory (LSTM) networks, as a more advanced form of GRU, are frequently combined with CNN for battery SOH estimation, owing to their enhanced ability to capture long-term temporal dependencies in time-series signals and to improve model generalization [21]. CNN-LSTM architectures may still suffer from unstable training dynamics and sensitivity to hyperparameter settings, which can degrade prediction consistency across datasets. Ouyang et al. [22] proposed a method for SOH estimation of lithium-ion batteries combining meta-learning and CNN-LSTM to improve the generalization ability of neural networks to different battery data and reduce the dependence on a large amount of target battery data. Despite the improved adaptability, the meta-learning framework introduces high computational complexity and demands careful task selection, which limits its practicality in real-time BMS applications. In addition, the fusion of LSTM with other models shows excellent performance, LSTM is often paired with other models due to its unique gating mechanism that can dynamically focus on key features. Pepe et al. [23] screened 68-dimensional features from charge-discharge voltage profiles by automatic feature extraction, combined with LSTM and GRU for SOH estimation, which greatly improved model generalization. The feature screening process is computationally intensive, and redundant features may still exist, potentially leading to overfitting when data diversity is low. Ren et al. [24] screened nine highly correlated features by

Pearson's correlation coefficient, and optimized LSTM hyperparameters by combining dynamic adaptive cuckoo search algorithm, which solved the problems of noise sensitivity and hyperparameter optimization in traditional LSTM models. However, this approach overlooks degradation indicators that are less strongly correlated but still possess informational value. Xu et al. [25] converted the charging and discharging curves into graph-structured data, extracted the spatial features using graph convolution network, and LSTM captured the temporal dependencies, which substantially improved the SOH estimation accuracy. The graph-based framework increases model complexity and requires careful graph construction, making it less suitable for lightweight BMS deployment. In the context of SOH estimation using data-driven techniques, it has been empirically validated that the accuracy of predictions is strongly influenced by the quality of the input variables. The input features, typically designated as health indicators (HIs) [26], and the correlation of HIs with SOH is assessed by the Pearson correlation coefficient (PCC) [27]. Experiments have shown that substandard HI have been found to introduce significant measurement disturbances and redundant information, posing challenges to the precision and robustness of data-driven SOH estimation. However, although HIs with high PCC is very useful for improving the accuracy of SOH estimation, HIs with low PCC also contains potential capacity degradation information. And due to the differences in the correlation between the same HIs and SOH in different datasets, the generalizability of the model can be greatly affected, there are fewer studies in this area. Therefore, ensuring that the model can handle HIs with high PCC and mining the nonlinear relationship between HIs and SOH with low PCC while ensuring excellent generalization of the model is a promising direction.

HIs can be extracted manually or automatically, Tang et al. [28] effectively solved the feature extraction problem of weak aging signals in the early cycle of lithium-ion batteries by automatically extracting spatial geometric features with CNN and capturing time-dependent features with LSTM. Zhao et al. [29] used a multi-branch transformer structure to automatically extract high-dimensional global features for early and accurate prediction of lithium-ion battery life. However, automatic feature extraction may overlearn the noise and reduce the model generalization ability when the amount of data is small, so manual feature extraction is preferred by researchers [30,31]. Lin et al. [32] extracted three health indicators within the 0.1 V voltage interval in the charging curve, and combined k-nearest neighbor classification (KNN) and support vector regression (SVR) for co-training to realize high-precision online estimation of SOH. Interestingly, a new neural network algorithm, Kolmogorov-Arnold networks (KAN) [33], was applied to SOH estimation. Unlike traditional multi-layer perceptrons (MLP), the KAN enables improved interpretability by structuring its prediction process through univariate function decomposition. This allows for a more transparent understanding of how nonlinear dependencies between HIs and SOH are learned in lithium battery aging. Zhang et al. [34] designed a composite model that integrates the KAN for extracting nonlinear features with LSTM networks for capturing time-series patterns, aiming to improve SOH estimation accuracy. To eliminate the dependency on full charge-discharge data, Peng et al. [35] designed a KAN-based SOH estimation framework using incremental capacity (IC) analysis to improve both efficiency and accuracy. Chen et al. [36] proposed a hybrid framework fusing improved U-Net and KAN for balancing local temporal features with global dependencies. Using KAN as the final prediction module, complex high-dimensional relationships are modeled by combinations of learnable univariate functions, which fill the deficiency of traditional U-Net in modeling global temporal dependencies. Most of the literature has focused on KAN's ability to characterize nonlinear relationships, but all have been features that are highly correlated with SOH, with little mention of using KAN for features with less high PCC, and studies of its generalizability.

To extract latent capacity degradation patterns from features exhibiting low correlation with the SOH, explore the effect of HIs with low correlation on the estimated generalizability of SOH, this study

proposes a dual-path perception framework that combines a Kolmogorov-Arnold Network-transformer (KAT) with CNN-LSTM architecture. By leveraging the unique B-spline curve design of the KAN within the KAT component and embedding KAN into the transformer architecture, the proposed model is able to extract hidden states from HIs and identify critical inflection points in the capacity degradation process. This design addresses the limitations of conventional attention mechanisms in capturing complex nonlinear relationships and integrating domain knowledge, thereby enhancing the generalization capability of the model. In addition, local features are extracted through the CNN module, while the LSTM component is employed to address the long-term dependencies in capacity degradation and to track the overall degradation trend. The HIs are divided into three gradients in this study, the KAT is applied to the HIs in the second and third gradients, while the CNN-LSTM is responsible for processing the HIs in the first gradients. Finally, the outputs of KAT and CNN-LSTM are adaptively weighted and summed up to obtain the accurate SOH. Furthermore, to highlight the effectiveness of the proposed method, CNN-LSTM is selected as a baseline model to emphasize the superior capability of the KAT component in mining latent capacity degradation information. MLP is also included as a baseline, as the KAN architecture is developed based on the MLP framework, which underscores the methodological advancement and innovation of the proposed approach.

The key original contributions of this work are summarized below:

- 1) A model capable of extracting latent capacity degradation information from HIs is developed, and the impact of HIs quality on SOH estimation is thoroughly examined. The problem of low accuracy of SOH estimation due to low mass HIs is solved, and the potential application of the model in the field of battery SOH estimation is verified.
- 2) Features are innovatively classified based on the degree of correlation with SOH, and the impact of HIs quality and quantity on SOH estimation is thoroughly investigated to determine the optimal

model input configuration, providing valuable insights for further research in battery SOH estimation.

- 3) Ablation studies are conducted to validate the model's advancement and offer information about its practical application for SOH estimation. The proposed method is further evaluated on three different battery datasets, demonstrating generalization and accuracy of the model across different cell types, with cross-dataset experiments confirming robust performance under varying temperatures, C-rates, and measurement platforms.

2. Battery modeling

2.1. Kolmogorov-Arnold networks-transformer

KAN originates from the Kolmogorov-Arnold representation theorem, and differs from the traditional multi-layer perceptron (MLP) in that KAN turns the weight parameters into a learnable B-spline, moves the activation function from nodes to edges, and the original nodes are simplified to a linear summation. The unique design of the B-spline not only embeds physical constraints in the KAN, but also taps into the nonlinear relationship between HIs and SOH.

As the degradation of batteries is a prolonged process, it is possible for abrupt capacity drops to occur at specific points throughout the aging cycle, this study introduces a transformer Encoder, to capture the critical stages of the aging process through the mechanism of self-attention with the structure of the is shown in Fig. 1, the model inference process is as follows.

First, the Kolmogorov-Arnold representation theorem can be summarized as follows:

$$f(X) = f(x_1, \dots, x_n) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^n \phi_{q,p}(x_p) \right) \quad (1)$$

where both $\phi_{q,p}$ and Φ_q all denote univariate functions in mathematics, which need to be applied to specific SOH estimations, and the

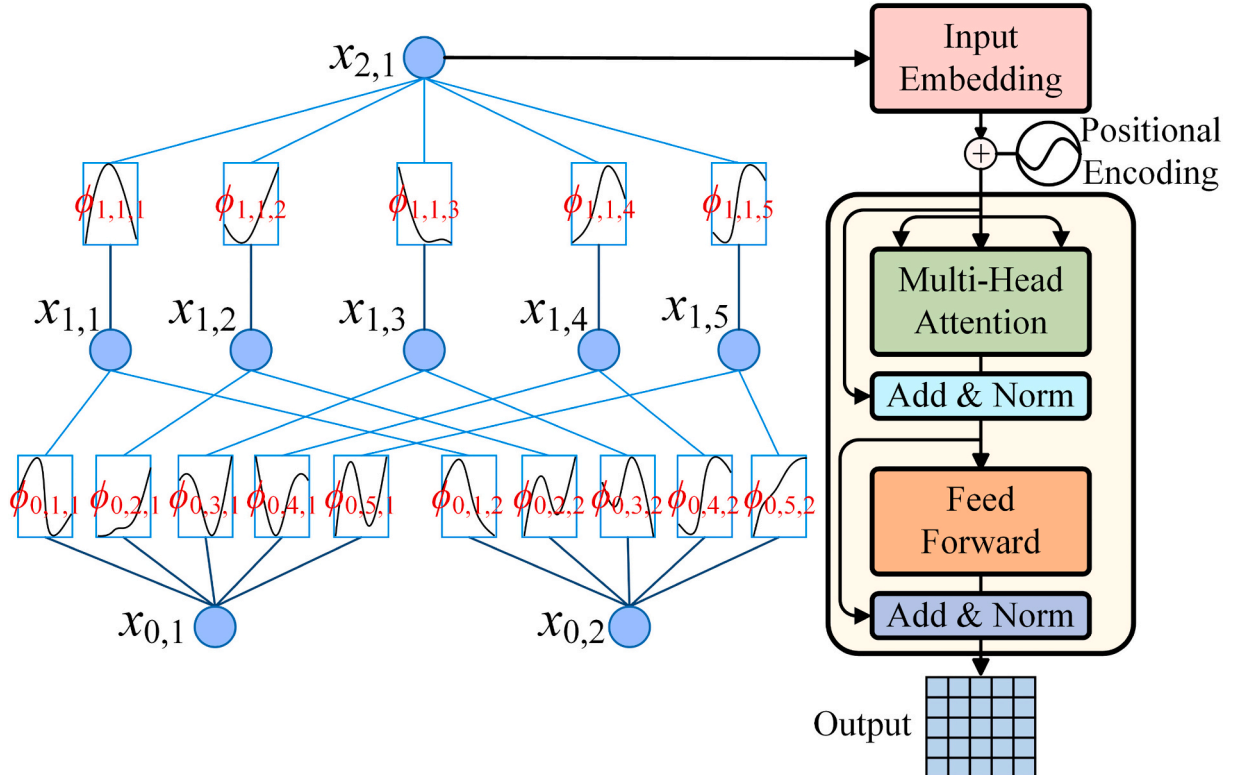


Fig. 1. Structure of the KAT.

conversion process is as follows (it is clearer when combined with Fig. 1), with reference to [33] for a complete derivation.

KAN can be simply represented as an array:

$$[n_0, n_1, \dots, n_L] \quad (2)$$

where n_i means the number of nodes in the i -th layer. In the study, i is set to 2, providing sufficient capability to model nonlinear mappings between health indicators and SOH while avoiding overfitting.

In other words, KAN is a stack of individual KANLinear, each of which can be represented as a one-dimensional matrix:

$$\Phi = \{\phi_{q,p}\}, \quad p = 1, 2, \dots, n_m, \quad q = 1, 2, \dots, n_{out} \quad (3)$$

where $\phi_{q,p}$ denotes the internal activation function and Φ_q means the external function, the activation function is responsible for mapping the input features X to the intermediate variables, and the external function is responsible for mapping linear combinations of the intermediate variables to the final output.

The activation function is expressed as:

$$\phi_{l,j,i}, \quad l = 0, \dots, L-1, \quad i = 1, \dots, n_l, \quad j = 1, \dots, n_{l+1} \quad (4)$$

where (l, i) denotes the i -th neuron in layer l , $(l+1, j)$ denotes the j -th neuron in layer $l+1$, and a total of $n_l n_{l+1}$ activation functions are employed between the l^{th} and $(l+1)^{th}$ layers.

The activation function between layers l and $l+1$ is represented as a matrix:

$$\phi(x) = \omega_b b(x) + \omega_s S(x) \quad (5)$$

where ω_b is for the trainable coefficients, optimized by backpropagation, $b(x)$ is the residual function, and ω_s is the learnable weight to mitigate local bias in the spline fit. $S(x)$ denotes the B-spline basis function defined as follows:

$$S(x) = \sum_i c_i B_i(x) \quad (6)$$

After going through the activation function layer, the node will perform a linear summation of the output with the following equation:

$$x_{l+1,j} = \sum_{i=1}^{n_l} \phi_{l,j,i}(x_{l,i}), \quad j = 1, \dots, n_{l+1} \quad (7)$$

Finally, combined with Eq. (2), the KAN model output is:

$$\text{KAN}(X) = (\Phi_{L-1} \circ \Phi_{L-2} \circ \dots \circ \Phi_1 \circ \Phi_0)X \quad (8)$$

Rewrite Eq. (8) in a form similar to Eq. (1):

$$f(X) = \sum_{i_{L-1}=1}^{n_{L-1}} \phi_{L-1,i_L,i_{L-1}} \left(\sum_{i_{L-2}=1}^{n_{L-2}} \dots \left(\sum_{i_2=1}^{n_2} \phi_{2,i_3,i_2} \left(\sum_{i_1=1}^{n_1} \phi_{1,i_2,i_1} \left(\sum_{i_0=1}^{n_0} \phi_{0,i_1,i_0} (x_{i_0}) \right) \right) \right) \dots \right) \quad (9)$$

The two features with lower PCC are passed into KAN after passing through the Encoder of transformer to output SOH, the Encoder formula is as follows.

First, in order to prevent the KAN output from disrupting the time series order after entering the transformer, it is necessary to generate the position code for the KAN output:

$$PE_{(\text{pos}, 2i)} = \sin\left(\frac{\text{pos}}{10000^{2i/d}}\right), \quad PE_{(\text{pos}, 2i+1)} = \cos\left(\frac{\text{pos}}{10000^{2i/d}}\right) \quad (10)$$

where pos indicates the location index, and i specifies the dimension in

the feature vector.

The next step will be the most important step in the transformer, which introduces the Query, Key, and Value matrices to compute the correlation between the elements in the sequence through the attention mechanism:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (11)$$

where d_k denotes the dimension of the vector, softmax is a weight normalization function that assigns a correlation weight to Q , K , and V respectively and ensures that the weights sum to 1.

The point of this study is to implement the nonlinear mapping in the traditional transformer through KANLinear, instead of using a single linear layer. The formulation is shown below:

$$Q = \text{KAN}_Q(X), K = \text{KAN}_K(X), V = \text{KAN}_V(X) \quad (12)$$

Mapping inputs to multiple subspaces is computed in parallel and then stitched:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O \quad (13)$$

where $\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$. W^O is the output projection matrix. In this study, $h = 3$.

Apply residual linking and layer normalization to multi-attention outputs:

$$Z = \text{LayerNorm}(X' + \text{MultiHead}(Q, K, V)) \quad (14)$$

Nonlinear transformation of the normalized vector:

$$\text{FFN}(Z) = \max(0, ZW_1 + b_1)W_2 + b_2 \quad (15)$$

The output of the encoder layer is produced by reapplying a residual connection and subsequently performing layer normalization:

$$X_{\text{out}} = \text{LayerNorm}(Z + \text{FFN}(Z)) \quad (16)$$

2.2. Convolutional neural network

CNN are often used in the field of battery estimation to extract localized features and capture time dependencies. The fundamental structure of a CNN comprises convolution, pooling, and fully connected layers. Fig. 2 provides a visual representation of this configuration. In a typical CNN pipeline, local feature extraction is conducted by the convolutional layer using weight-shared filters, and nonlinear mappings to SOH are achieved through activation functions. To prevent overfitting, a pooling layer is applied, and the final output is produced by the fully connected layer. In this study, the fully connected layer is not used in the

CNN because the CNN is to be used in tandem with the LSTM.

The CNN formulation is as follows:

$$y_{i,j} = \sum_{m=0}^{k-1} \sum_{n=0}^{k-1} x_{i+m,j+n} \omega_{m,n} + b \quad (17)$$

where k denotes the convolution kernel size and b is the bias term. In this study, k is set to 3×3 and the step size of the convolution kernel is 1, the number of convolutional layers is set to 3, which effectively extracts hierarchical spatial features while keeping the model complexity moderate.

The activation function formula is as follows:

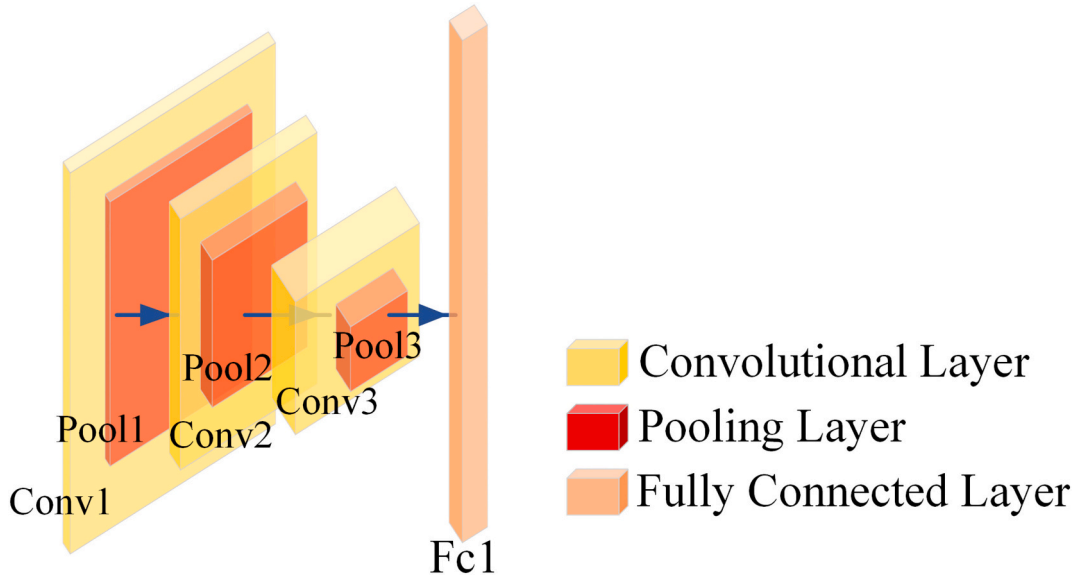


Fig. 2. Structure of the CNN.

$$C = \varphi(Y) \quad (18)$$

where Y represents the output of the convolutional layer and φ is the activation function.

The pooling layer is given by:

$$y = \text{maxpool}(C, p, s) \quad (19)$$

where C denotes the output of the activation function, p denotes the pooling size, which is set to 3×3 , and s denotes the pooling operation step size, which is set to 2 to reduce feature map dimensions while retaining important information.

2.3. Long short-term memory network

To overcome the vanishing gradient issue in handling long sequences, LSTM extends the basic RNN framework by embedding memory components. Functionally, it can be regarded as an enhanced version of both RNN and GRU. The LSTM network is effective in learning long-range temporal patterns, making it a favorable choice for modeling capacity fading in lithium-ion batteries over prolonged charge-discharge processes. Fig. 3 shows the structure of LSTM. To maintain and update temporal information effectively, LSTM introduces a core memory cell (C_t), along with three key nonlinear gating units: the input gate, forget gate, and output gate, which collaboratively control the data flow and

retention.

Next, the three gating units in LSTM will be introduced. The forgetting gate is responsible for deciding which features from the previous moment need to be retained or discarded, and outputs a probability value from 0 to 1 through an activation function with the following formula:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (20)$$

where W_f is the forgetting weight matrix, h_{t-1} contains historical battery aging information, x_t is the current input feature, b_f is the bias term, and σ is the activation function.

The input gate consists of an input screening unit and a candidate memory unit, which are used for the update level of the current input features and for extracting some of the aged features, respectively, with the following formulas:

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (21)$$

$$\tilde{C}_t = \tanh(W_C[h_{t-1}, x_t] + b_C) \quad (22)$$

where W_i is the weight matrix of the input screening unit that controls the update of the current features and b_i is the bias of the input gate.

The results of the forgetting gate and the input gate will be used in the state update unit for integrating the historical aging characteristics

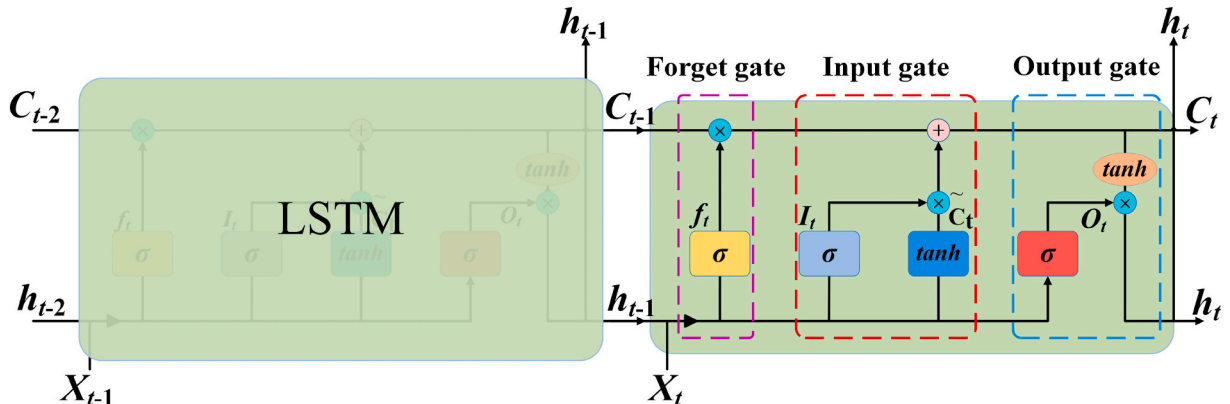


Fig. 3. Structure of the LSTM.

and the current aging characteristics of the battery with the following equation:

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad (23)$$

where $\odot$ denotes the multiplication of the principal elements.

The output gate screens the current state information for use in the final output with the following formula:

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (24)$$

$$h_t = o_t \odot \tanh(C_t) \quad (25)$$

where the output gate parameters include the weight matrix W_o and bias b_o , and the hidden state h_t captures both current inputs and temporal aging characteristics.

2.4. Experimental design and datasets

In this study, two public datasets and one experimental dataset were used for simulation, the public datasets were obtained from Center for Advanced Life Cycle Engineering (CALCE) and Sandia National Laboratories (SNL) for testing and validation, respectively, and the experimental dataset will also be used to validation, which is presented in Section 4.

Batteries used in the CALCE dataset feature a nominal capacity of 1.1 Ah, with LiCoO_2 serving as the anode component. Under an ambient temperature of 25 °C, all batteries were subjected to identical charge-discharge procedures: charging began with a 0.5C constant current until the voltage reached 4.2 V, transitioned to a constant voltage phase until the current fell below 0.05 A, and was followed by discharge down to a cutoff voltage of 2.7 V.

The SNL dataset comprises 6 sets of data, employing different

materials in cyclic testing experiments while setting varying discharge rates and ambient temperatures. As one of the validation datasets, the dataset with negative electrode material LiFePO_4 , rated capacity of 1.1 Ah, ambient temperatures of 15 °C, 25 °C, and 35 °C, and discharging rates of 1C and 2C, respectively, were selected for the validation experiments of the model generalization in this study.

To validate the effectiveness of the proposed method, cyclic charge-discharge experiments were conducted under laboratory conditions. The experimental setup, illustrated in Fig. 4, consisted of a lithium-ion battery cell, a battery testing system, a thermal chamber, an electrochemical workstation, and a host computer. The lithium-ion cell was assembled using 16 identical 18,650 cells (Shenzhen Zhuoneng), each with a rated capacity of 2.7 Ah and $\text{Li}(\text{NiCoMn})\text{O}_2$ as the cathode material. The NEWARE CT-400 system was employed for battery charge and discharge operations, with a discharge current set at 1890 mA. The thermal chamber maintained the environment at room temperature. An electrochemical workstation measured the battery current, while the host computer was used for data acquisition and storage.

3. State estimation algorithm

3.1. Parallel modeling framework

In this research, a parallel CNN-LSTM and KAT structure is adopted to predict battery SOH. The approach innovatively integrates KAT for extracting information from low-correlation features, employs KAN to improve nonlinear modeling performance, and embeds domain-specific knowledge into the transformer using KANLinear to guide attention distribution. This facilitates the identification of critical degradation trends and improves the recognition of subtle feature information. Meanwhile, CNN-LSTM is employed to deal with a highly correlated HIs which contains explicit health degradation patterns to address long-term

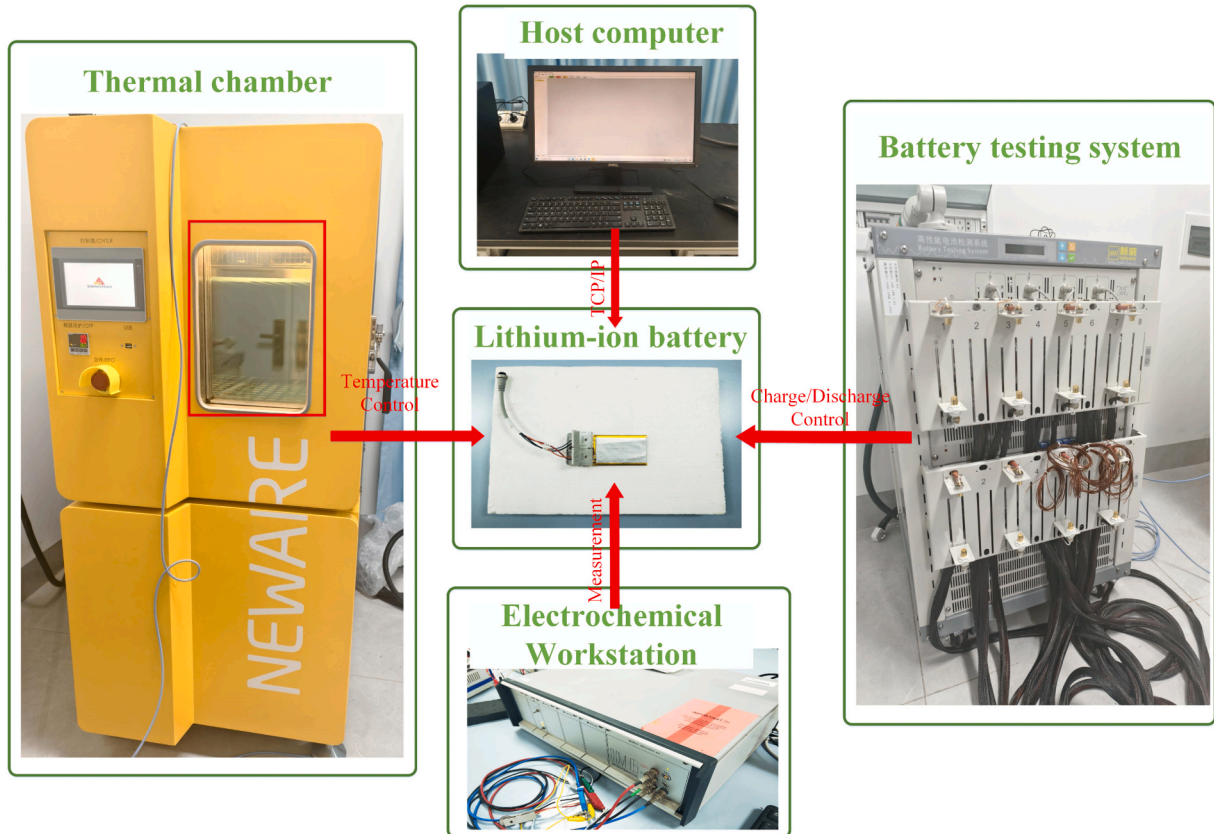


Fig. 4. Test bench.

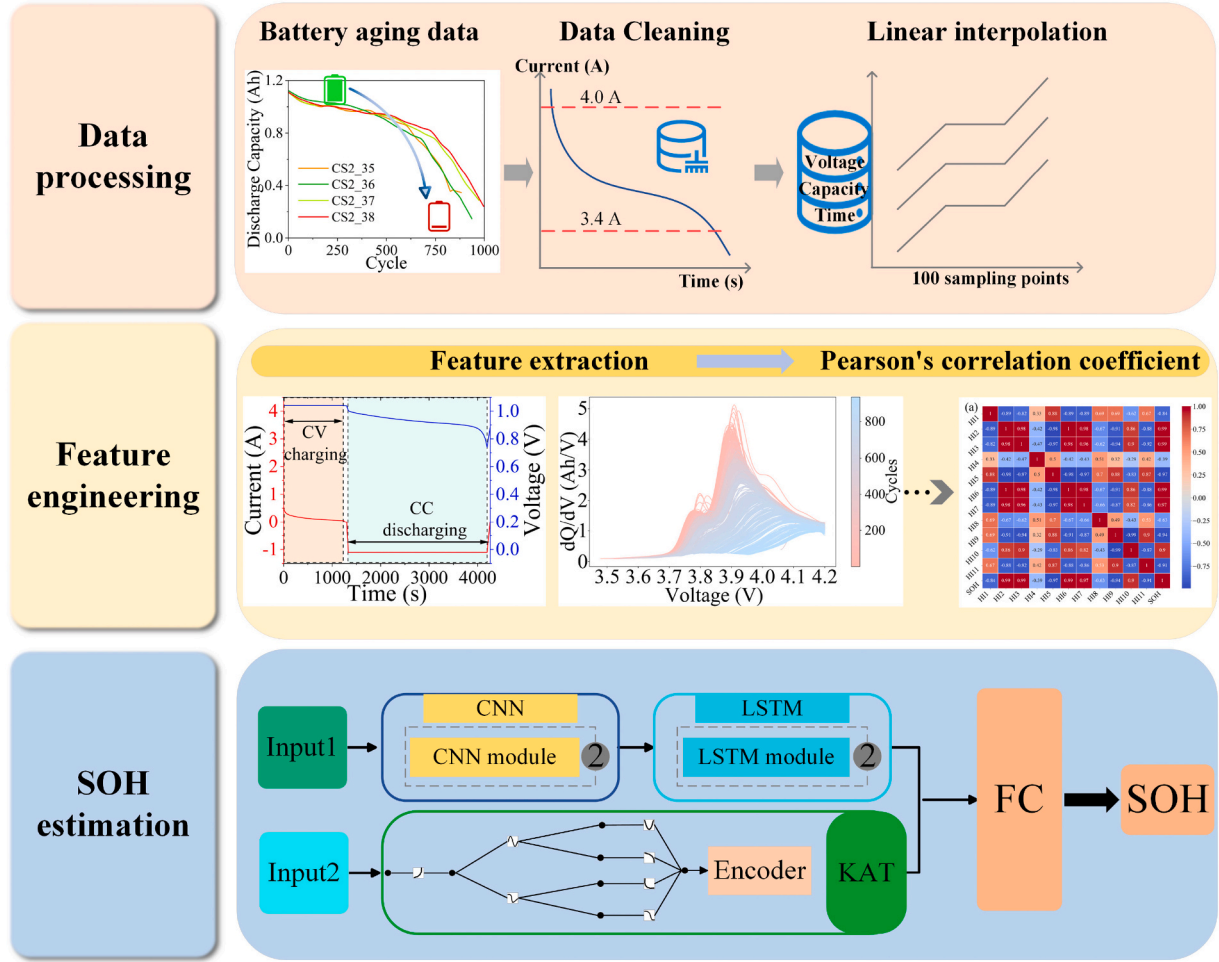


Fig. 5. Overall framework of the paper.

Table 1
Algorithmic procedure for dual-path estimation of SOH.

Algorithm	Procedure
KAT	Initialization: Input-output pairs: $D_f: \{x_i, y_i\}$ (1) Input processing: $X = \text{KAN}(x_i, \text{update_grid})$ (26) (2) Generating attention parameters: $\begin{cases} Q = \text{Query}(X) \\ K = \text{Key}(X) \\ V = \text{Value}(X) \end{cases}$ (27) (3) Auto-attentional coding and getting the output. (4) The output $\hat{y}_1$ of KAT is obtained by dimensionality reduction of the fully connected layer.
CNN-LSTM	Initialization: Input-output pairs: $D_f: \{x_j, y_j\}$ (1) Convolutional features are extracted and upsampled to obtain. $H_1 = \phi(w_1 X + b_1)$ (28) $H_2 = \phi(w_2 H_1 + b_2)$ (29) (2) After the LSTM component to get the output. (3) Full connectivity layer dimensionality reduction to get the output $\hat{y}_2$.
Adaptive weighted fusion	(1) Weight normalization: $\begin{cases} \omega_1 = \sigma(w_{\text{KAT}}) \\ \omega_2 = \sigma(w_{\text{CNN_LSTM}}) \end{cases}$ (30) (2) Dynamic weighted summation: $\hat{y} = \frac{\omega_1}{\omega_1 + \omega_2} \hat{y}_1 + \frac{\omega_2}{\omega_1 + \omega_2} \hat{y}_2$ (31)

temporal dependence with local mutation. Finally, the outputs of KAT and CNN-LSTM are weighted and summed to obtain the exact SOH using

dynamic adaptive weighted summation in the fully connected layer. Fig. 5 outlines the research workflow adopted in this study, and Table 1 summarizes the key components of the model estimation framework.

3.2. Feature engineering

Feature engineering is the most basic part of SOH estimation, where different HIs are extracted by certain computational rules. Many researchers choose the data of constant current or constant voltage stage of charge-discharge for feature extraction, and generally take the average value to calculate the features. During this stage, both current and voltage remain stable, providing favorable conditions for reliable feature quantification. Most of the literature considers HIs with PCC greater than 0.8 as excellent HIs when performing SOH estimation, and some of them regard HIs with PCC greater than 0.7 as excellent HIs. A comparative study of different numbers of HIs inputs reveals that too many HIs introduce more noise, which leads to an accumulation of SOH estimation errors when the model's noise immunity is low. Whereas a smaller number of HIs will result in the model not being able to learn the accurate capacity recession law, leading to a larger SOH estimation error. Fig. 6(a) and (b) shows the Pearson correlation analysis of features extracted from the CALCE and SNL datasets, respectively. In this study, three HIs were found to be the best for the SOH estimation accuracy of the proposed method, so the model inputs in this study were all set to 3.

In this study, HIs is divided into three gradients based on PCC values. The first gradient represents average discharge voltages where PCC exceeds 0.7, exhibiting strong linear correlation with capacity decay and reflecting electrochemical aging characteristics. The second gradient

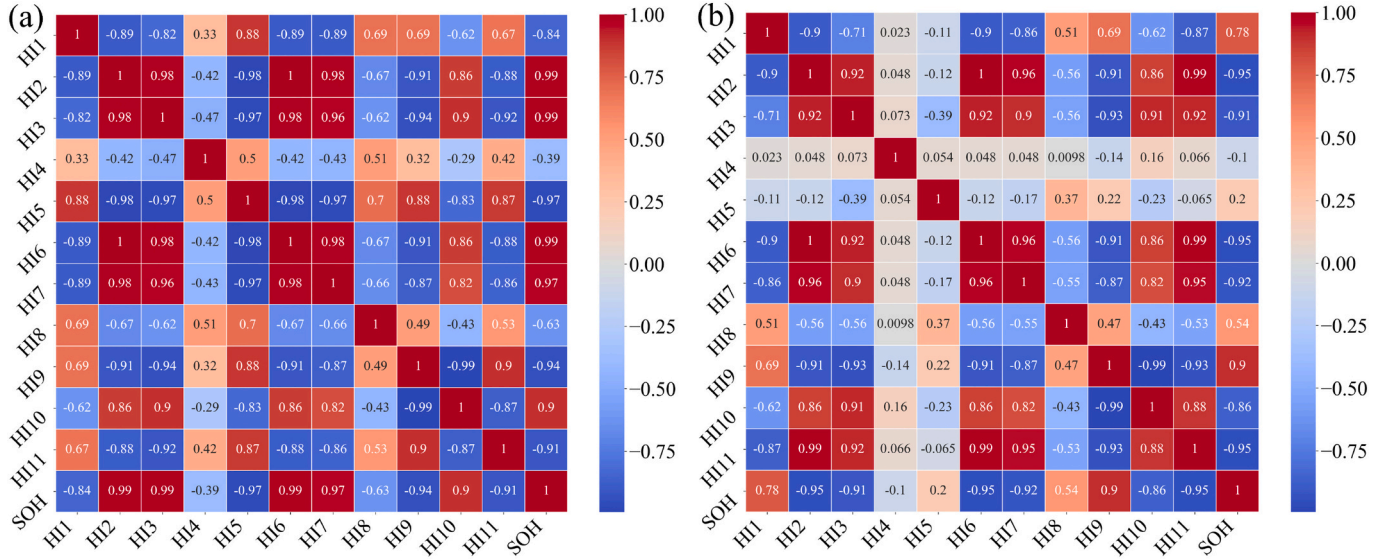


Fig. 6. Analysis of Pearson's correlation coefficient. (a) A sample of the CALCE dataset. (b) A sample of the SNL dataset.

represents the charge-to-discharge time ratio with PCC values between 0.4 and 0.7. This metric captures the battery's internal dynamic decay characteristics and identifies nonlinear decay information related to internal resistance and rate performance. The third gradient is the pre-charge initial voltage with PCC below 0.4, which provides subtle decay clues under low-correlation conditions, thereby enhancing model robustness and generalization capability. All HIs in this study utilize these three features, the feature calculation methods in the three gradients are shown as (32)–(34).

$$V_{avg} = \frac{1}{N} \sum_{i=1}^N V_i^{discharge} \quad (32)$$

$$R_t = \frac{t_{charge}}{t_{discharge}} \quad (33)$$

$$V_{init} = V(t_0) \quad (34)$$

Although the feature extraction process is consistent with traditional methods, the innovation of this study lies in the utilization of these features. Traditional methods typically only input features with high correlation coefficients into a single prediction model, disregarding those with low correlations, and thus neglecting low-correlation but physically meaningful degradation signals. In contrast, the proposed method assigns HIs of different correlation degrees to different network components: features with high and medium correlation coefficients are input into the CNN-LSTM module to capture explicit spatio-temporal

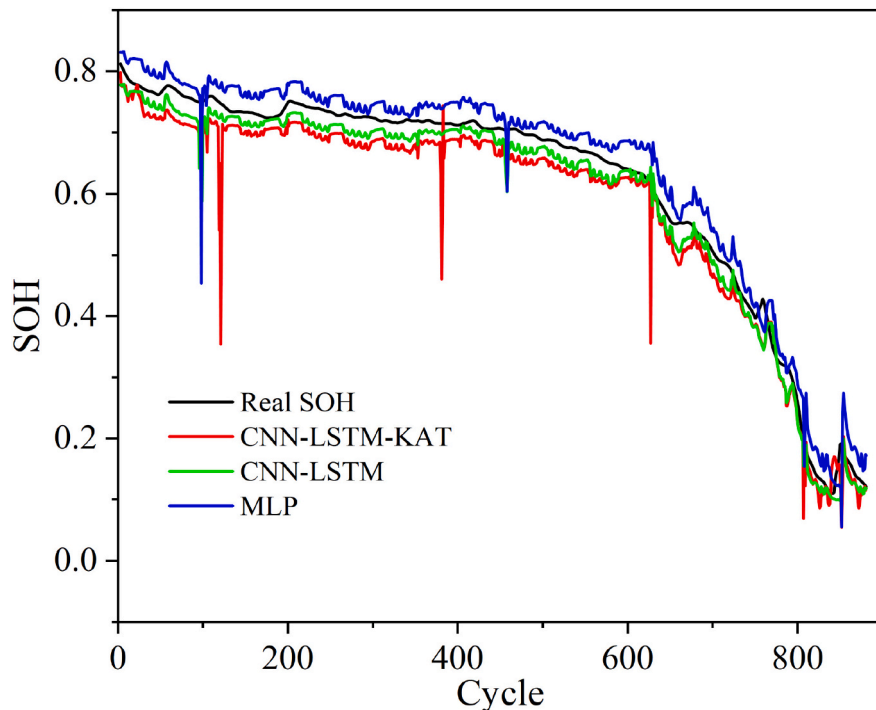


Fig. 7. Estimation results for the CS2_35 dataset before undifferentiated HIs gradients.

Table 2

SOH estimation results before input features are adjusted.

Before	Model	RMSE (%)	MAE (%)
CS2_35	CAFM	10.25	9.76
	CNN-LSTM	7.81	7.52
	MLP	8.05	7.70
CS2_36	CAFM	10.84	10.06
	CNN-LSTM	7.27	6.84
	MLP	8.00	7.43
CS2_37	CAFM	7.37	7.30
	CNN-LSTM	5.34	5.20
	MLP	5.74	5.23
CS2_38	CAFM	12.16	10.91
	CNN-LSTM	8.86	8.07
	MLP	8.94	8.51

degradation patterns, while features with low correlation coefficients are processed by the KAT module, which is adept at mining nonlinear and latent degradation information. This gradient-based feature allocation enables comprehensive utilization of both strong and weak degradation cues, effectively alleviating the inherent information loss in the traditional single-path feature processing strategy.

3.3. SOH estimation

In this study, the HIs of the three gradients is simultaneously transmitted to CNN-LSTM and KAT, respectively, and then adaptive weighted summation is performed to obtain the SOH. Fig. 7 demonstrates the estimation results of CS2_35, and Table 2 shows the SOH estimation results obtained by the model receiving 3 gradients of HIs inputs. In order to emphasize the advancement of the proposed method, model ablation experiments are added in this study, which are CNN-LSTM and MLP, respectively. The reason for choosing MLP for the ablation experiments is that KAN is developed on the basis of MLP, which can show that KAN is superior to the MLP model and at the same time highlights its ability to mine the hidden states. In addition, it should be noted that at this point in time, the inputs to both the model in this study and the ablation experiment are three identical HIs. However, after distinguishing the gradients of the HIs received by the models in this study, the HIs received by both CNN-LSTM and MLP are in the first gradient, i. e., the PCC is greater than 0.7. This is primarily because the model inputs corresponding to the three gradients contain considerable noise, while the second and third gradients provide limited information regarding capacity degradation. As a result, the estimation performance of models based on these gradients is less reliable. Therefore, three HIs from the first gradient were selected to more effectively demonstrate the advantages of the proposed method.

The results indicate that the performance of the proposed model improves when it simultaneously receives all three gradients of the HIs as input, the results are much worse than the ablation experimental model, with poor robustness and constant SOH fluctuations. This occurs because the CAFM model employs a parallel feature pathway architecture, where the CNN-LSTM branch extracts temporal and local degenerate patterns, while the KAT branch captures correlations across feature dimensions. When the model incorporates features with three gradients simultaneously, the KAT module amplifies noise in weakly correlated features while capturing complex relationships, leading to abrupt short-term fluctuations in feature interaction weights. This “correlation conflict” effect causes the model to produce instantaneous prediction mutations at several cycle points, manifesting as sharp fluctuations in the curve. Although the CNN-LSTM model also handles mixed feature inputs, its structure primarily relies on time-series feature extraction and exhibits lower sensitivity to feature correlations. Consequently, it can partially smooth out disturbances caused by noise. The MLP model lacks temporal modeling capabilities, directly mapping

features to SOH outputs, and thus cannot effectively distinguish between degradation trends and random disturbances. Its predictions typically manifest as discrete fluctuation points, appearing as abnormal deviations at specific cycles. Although the amplitude of individual fluctuations is relatively limited, the overall curve exhibits poor stability.

After CNN-LSTM processes the HIs from the first gradient and KAT receives those from the second and third gradients, the input complexity of the model is significantly reduced, allowing KAT to more effectively capture and refine the latent representations within the HIs. The SOH estimation results for the CALCE dataset are displayed in Fig. 8, including CS2_35, 36, 37, 38, where the left part is the SOH estimation comparison plot, the right part is the SOH estimation error violin plot. A box plot forms the core of the error visualization, with the horizontal line used to indicate the mean estimation error, and the “×” symbol marking the median. The white box illustrates the interquartile range (IQR), spanning from the 25th to the 75th percentile, thereby encompassing the middle 50 % of the error values. Surrounding the box plot is a kernel density estimation curve, which depicts the relative frequency of error values within the overall distribution. A broader distribution indicates a higher concentration of values in that range. Table 3 demonstrates the SOH estimation metrics after differentiating the gradients. The root mean square error (RMSE) and mean absolute error (MAE) are employed to quantify the accuracy of the overall SOH estimation, while the coefficient of determination (R^2) is used to evaluate the model's fit, as shown in Fig. 9. The mathematical expressions of these metrics are listed below.

$$\text{RMSE} = \sqrt{\frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i)^2} \quad (35)$$

$$\text{MAE} = \frac{1}{m} \sum_{i=1}^m |y_i - \hat{y}_i| \quad (36)$$

$$R^2 = 1 - \frac{\sum_{i=1}^m (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^m (\bar{y}_i - \bar{y})^2} \quad (37)$$

where y_i is the measured SOH value, $\hat{y}_i$ is the predicted SOH value, $\bar{y}$ is the average value of the measured SOH, and m is the number of cycles.

In order to further prove the advancement of the proposed method, after distinguishing the input gradients of the proposed model, the input of the comparison model is no longer the three HIs in the three gradients, but the three HIs in the first gradient, that is, the HIs with PCC greater than 0.7. As shown by the results, the proposed model more accurately predicts SOH values and exhibits smaller deviations from the measured data than the baseline approaches. In the four CALCE datasets, the proposed method performs the best in CS2_35, and CS2_38 is relatively the lowest, but its results are all better than the other models. In the four datasets, the MAE of the proposed method is on average more than 40 % lower than the other two models, and the accuracy is improved by about 32 % even though the RMSE is relatively sensitive to the results. As can be seen in the R^2 presented in Fig. 9, the proposed model generally fits better in the datasets, demonstrating the ability of the proposed method to not only learn the relationship between high-correlation HI and SOH, but also to tap the performance of low-correlation HI in the process of capacity degradation. Furthermore, despite integrating a CNN-LSTM with a KAT module, the proposed CAFM maintains computational efficiency within a feasible range for practical applications. CAFM's total parameter count is approximately 1.9×10^5 , only 1.6 times higher than the CNN-LSTM model. Benefiting from parallel computation and lightweight feature dimensions, its average inference time falls below 20 milliseconds per sample, meeting the stringent latency requirements of battery management systems.

To verify the effectiveness of the proposed method, we conducted a comparative evaluation on the CALCE dataset against baseline models

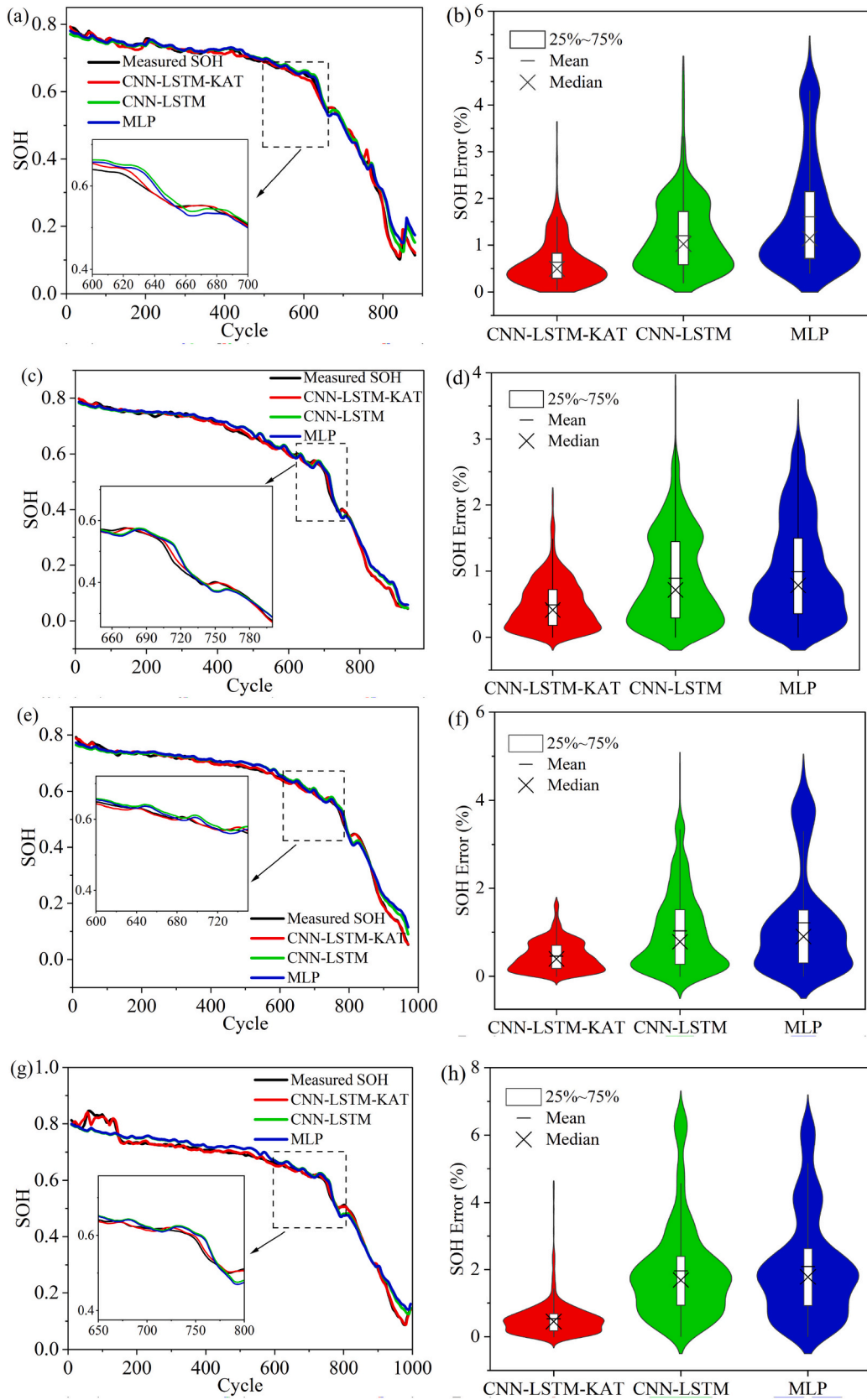


Fig. 8. Estimation results for the CALCE dataset. (a), (b) CS2_35. (c), (d) CS2_36. (e), (f) CS2_37. (g), (h) CS2_38.

Table 3

Comparison of SOH estimation results with baseline and SOTA models after input features are adjusted.

Battery	Model	RMSE (%)	MAE (%)	R^2
CS2_35	CAFM	0.78	0.59	0.9963
	CNN-LSTM	0.93	0.85	0.9131
	MLP	1.09	0.92	0.8801
	TVF-EMD-BiGRU	0.99	2.5	/
	Attention-based [39]	1.49	0.89	0.9949
	CNN-BiLSTM [40]			
CS2_36	CAFM	0.62	0.49	0.9835
	CNN-LSTM	0.92	0.72	0.9792
	MLP	1.05	0.94	0.9025
	TVF-EMD-BiGRU	1.24	3.1	/
	Attention-based [39]	2.13	1.16	0.9932
	CNN-BiLSTM [40]			
CS2_37	CAFM	0.57	0.46	0.9970
	CNN-LSTM	0.76	0.62	0.9370
	MLP	0.86	0.80	0.8743
	TVF-EMD-BiGRU	2.96	9.89	/
	Attention-based [39]	1.49	0.8	0.9948
	CNN-BiLSTM [40]			
CS2_38	CAFM	0.75	0.53	0.9931
	CNN-LSTM	1.23	1.04	0.8935
	MLP	1.29	1.02	0.8194
	PSO-BiLSTM-Attention	1.4	0.8	0.9707
	Attention-based [41]	1.78	1.5	0.9923
	CNN-BiLSTM [40]			

(MLP, CNN-LSTM) and recent state-of-the-art (SOTA) approaches reported in the literature. As summarized in Table 3, the proposed CAFM consistently achieves the best overall performance across all cells, with lower RMSE and MAE values and higher R^2 scores. Compared with the baseline CNN-LSTM model, CAFM reduces prediction errors by approximately 30–40 % on average, demonstrating stronger estimation accuracy and smoother trend tracking. In addition, CAFM either matches or surpasses the performance of published SOTA methods such as TVF-EMD-BiGRU and Attention-based CNN-BiLSTM, confirming its superior capability for accurate and robust SOH estimation.

4. Validation and analysis

4.1. SOH estimation

In the evaluation of the experimental dataset, the same ablation experiments are used for comparative analysis. As with the inputs to the CALCE dataset, the inputs to the proposed method are three of the three gradients HIs, the inputs to the ablation experiment are the three HIs in the first gradient, and the SOH estimation results are shown in Fig. 10. As shown in the figures, the proposed method exhibits excellent SOH fitting performance. While the CNN-LSTM model also yields estimates relatively close to the observed SOH, its overall accuracy is noticeably inferior to that of the proposed approach. The MLP model demonstrates the least accurate estimation results among the compared methods. As a more sensitive indicator of estimation error, RMSE reveals that the proposed model performs better than the other two methods, with its RMSE being 48 % and 56 % lower, respectively. The proposed method outperforms other models in SOH estimation accuracy, even though the input data it utilizes is of lower quality than that in the ablation experiments. This result underscores the CAFM's effectiveness in capturing the latent states associated with capacity degradation.

4.1.1. Validation under different training proportions

In the process of SOH estimation using deep learning, 80 % of the entire dataset is generally used as a training set for learning the relationship between features and targets, and 20 % as a test set for testing the model effect. To provide additional validation of the proposed method's advancement, SOH estimation performance is evaluated using varying proportions of the training data. Fig. 11(a) corresponds to a training set proportion of 60 %, while Fig. 11(b) represents a 40 % training set. The black curve denotes the measured SOH, whereas the colored curves indicate the SOH estimation results of different models on the test set. As shown in Table 4, the SOH estimation performance of the models is evaluated under different training set proportions using the laboratory dataset. Despite a reduction in training data, the CAFM maintains stable performance, with RMSE and MAE staying under 1 %, and its fitting capability remains notably strong. The other two models showed larger fluctuations, especially when the training set was 40 %,

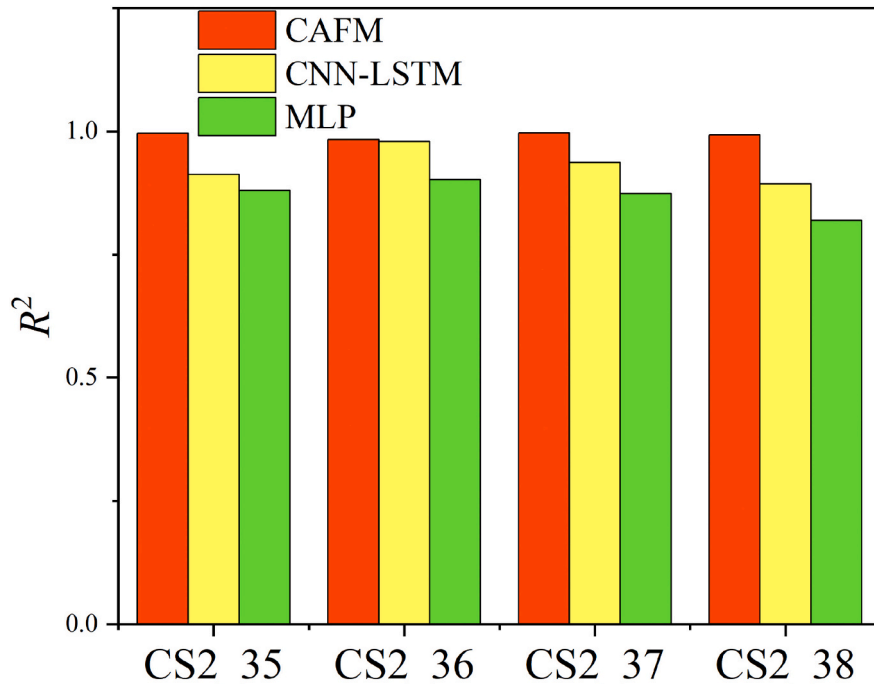


Fig. 9. R^2 for different models in the CALCE dataset.

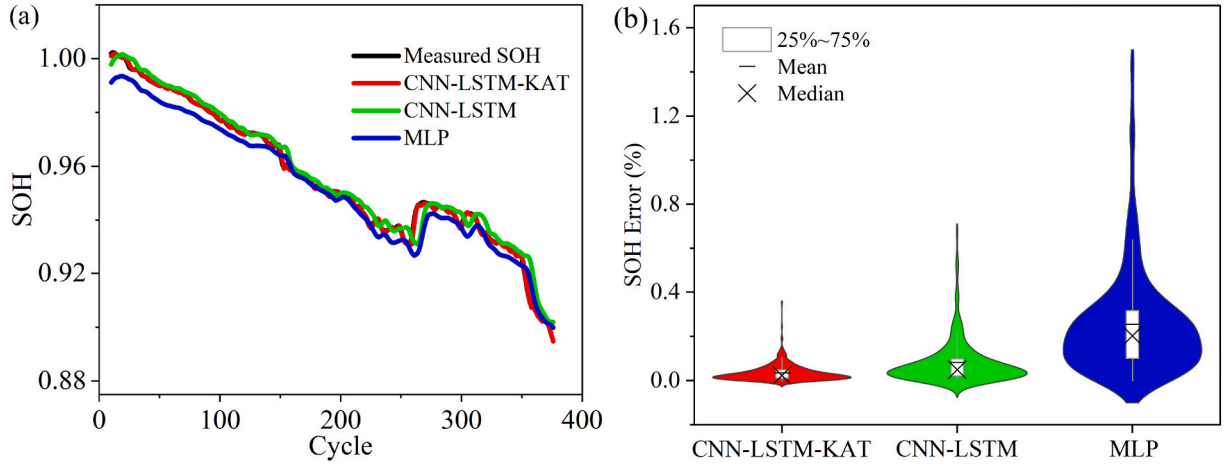


Fig. 10. Laboratory battery dataset estimation results.

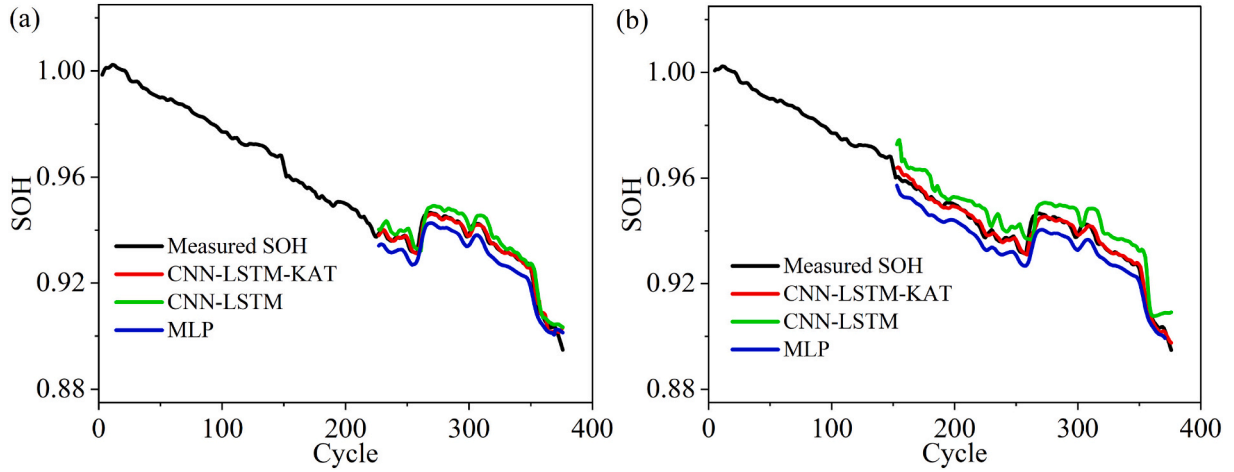


Fig. 11. (a), (b) SOH estimation results for the laboratory dataset at 60 %, 40 % training set.

Table 4
Laboratory battery SOH estimation indicator results.

Training set proportion (%)	Model	RMSE (%)	MAE (%)
80	CAFM	0.51	0.43
	CNN-LSTM	0.99	0.91
	MLP	1.17	1.01
60	CAFM	0.90	0.86
	CNN-LSTM	1.43	1.52
	MLP	1.82	1.79
40	CAFM	0.97	0.89
	CNN-LSTM	2.14	1.89
	MLP	2.76	2.56

and the RMSE exceeded 2 %. The CAFM is still able to generalize in small training set samples, indicating that the model avoids overfitting in SOH estimation.

4.2. Generalization verification

The SNL dataset used in this study includes multi-condition datasets at 1C/2C rates at 15 °C, 25 °C, and 35 °C. However, currently, few studies have conducted intra-domain SOH prediction or reported repeatable SOTA metrics on SNL subsets. Therefore, to ensure fairness and comparability, the SOTA of this study is relatively focused on the CALCE benchmark. For the SNL dataset, only the implemented baselines

(CNN-LSTM and MLP) are used for comparison, emphasizing the robustness and trend tracking ability of the model under different temperatures and charge-discharge rates. This method not only demonstrates its superior absolute performance in public benchmark tests, but also highlights its strong generalization and stability in cross-condition evaluation.

4.2.1. Results under 1C discharging rate

To further evaluate the model's generalization capability, experiments were conducted on the SNL dataset. This dataset contains lithium iron phosphate batteries, whose electrode composition differs from that of the laboratory dataset and CALCE dataset used previously. The dataset includes measurement data under various operating conditions, covering discharge rates of 1C and 2C, as well as three ambient temperatures: 15 °C, 25 °C, and 35 °C. Similarly, the proposed method utilizes three HIs from three gradient levels, while the comparative model is limited to three HIs with a PCC above 0.7.

First, a comparison of the models was made in the 1C discharging rate dataset, and Fig. 12 shows the SOH fit and error violin plots, with SOH estimates at 15 °C in Fig. 12(a) and (b), 25 °C in Fig. 12(c) and (d), and 35 °C in Fig. 12(e) and (f). Although the CNN-LSTM model is capable of broadly capturing the battery degradation trend, its accuracy in SOH estimation deteriorates progressively with rising temperature, whereas the performance of the MLP model is comparatively poorer. In addition, it can be seen that the prediction effects of both CNN-LSTM

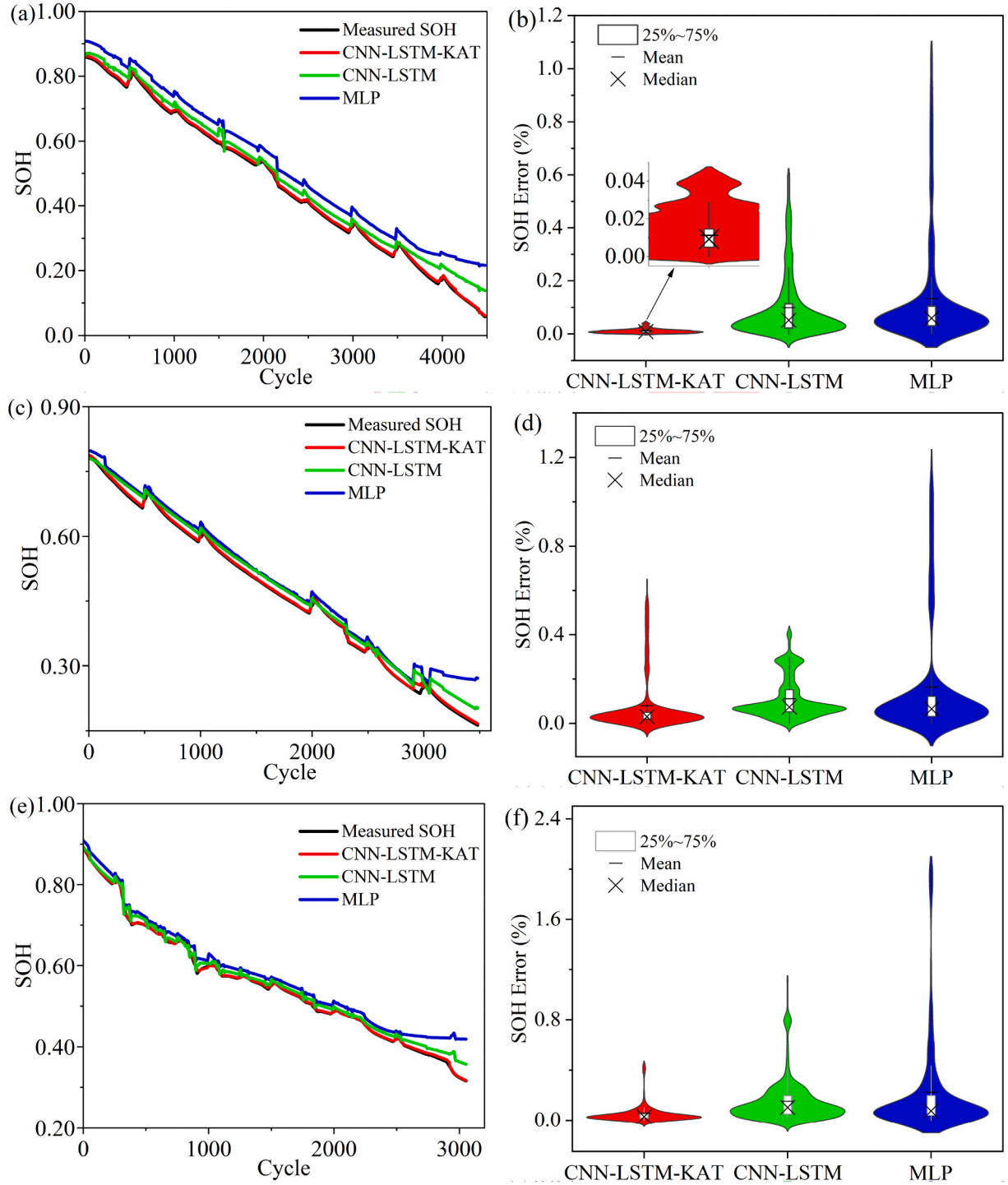


Fig. 12. SOH estimation results for 1C discharging rate at different temperatures. (a), (b) 15 °C. (c), (d) 25 °C. (e), (f) 35 °C.

and MLP models increasingly deviate from the capacity degradation trend in the later stages, especially at 25 °C, indicating that the models have limited generalization and a complete and reliable estimation of SOH cannot be easily achieved. On the contrary, the CAFM model still performs more stably in all three temperatures and is able to estimate SOH accurately, Table 5 shows the SOH estimation metrics of the models in all three temperatures at 1C discharging rate. Under the 1C discharge rate condition, the R^2 performance of the three models at different temperatures is illustrated in Fig. 13. Among them, the CAFM exhibits the highest fitting accuracy and overall consistency across temperature variations.

In the error distribution plots demonstrated in Fig. 12, the mean and median SOH estimation errors of the CNN-LSTM model are very close to each other, and the SOH fitting effect is second only to that of CAFM, which is effective for spatio-temporal modeling of high PCC features. CNN-LSTM model struggles to effectively learn hidden information characterized by low Pearson correlation, highlighting a limitation in its feature extraction capacity. The MLP model adopts a relatively basic architecture that applies multiple layers of nonlinear transformations to the input HIs, without the capacity to model complex degradation mechanisms, the estimation performance is highly susceptible to interference from high-noise features. This limitation results in a wide

Table 5
SOH estimation results for 1C discharging rate at different temperatures.

1C	Model	RMSE (%)	MAE (%)	R^2
15 °C	CAFM	0.33	0.26	0.9816
	CNN-LSTM	0.93	0.85	0.8942
	MLP	1.09	0.92	0.8437
25 °C	CAFM	0.29	0.23	0.9926
	CNN-LSTM	0.92	0.72	0.9201
	MLP	1.05	0.94	0.8491
35 °C	CAFM	0.56	0.37	0.9408
	CNN-LSTM	0.76	0.62	0.8637
	MLP	0.86	0.80	0.8317

distribution of estimation errors and a higher occurrence of outliers. The proposed method demonstrates optimal performance on the dataset. The mean SOH estimation error closely aligns with the median, indicating a symmetrical error distribution. The 25 %–75 % quantile interval is remarkably narrow, and the kernel density peaks within this range, suggesting that most of the model's SOH estimation errors fall within a small, consistent range. Furthermore, the maximum RMSE at 35 °C is only 0.56 %, and the R^2 value reaches 0.94, indicating that the model exhibits high predictive stability with no significant outliers.

4.2.2. Results under 2C discharging rate

Fig. 14 demonstrates the SOH estimation results of the battery at 2C discharging rate for three temperatures. Similar to the results at 1C, the estimation accuracy of MLP is the lowest among the three models, and CNN-LSTM can roughly estimate the capacity degradation trend, but its performance is much lower than that of CAFM. Table 6 shows the evaluation metrics of the three models. CAFM can still accurately estimate SOH at a discharging rate of 2C, and its RMSE and MAE are the lowest at 25 °C, 0.31 % and 0.38 %, respectively. As reflected by the R^2 scores in Fig. 15, the proposed approach outperforms all competing models in terms of SOH prediction accuracy.

In the results presented in Fig. 14, the early loops all closely track the measurement of SOH, and CNN-LSTM has some noise immunity, but in the middle and late loops the error increases significantly, and there is a lag at the inflection point of the capacity decay. MLP has the worst

performance ability, and the prediction of the inflection point is much less than that of CNN-LSTM. CAFM demonstrates consistent stability, with error distributions tightly centered and mean and median values in close agreement, highlighting its superior noise-tolerant characteristics. The proposed method not only tracks capacity degradation trends but also accurately captures SOH inflection points, which is attributed to KAT's dynamic weight allocation for low-PCC features that effectively suppresses hysteresis errors.

4.3. Cross-dataset generalization of SOH estimation

To comprehensively evaluate the robustness and direct-transferability of the proposed CAFM model, a cross-dataset generalization experiment was conducted. In practical applications, lithium-ion batteries often operate under diverse temperature and C-rate conditions, and data collected from different experimental platforms may exhibit domain shifts in distribution and measurement noise. Such discrepancies may considerably hinder model generalization, underscoring the importance of evaluating whether a model trained on a specific dataset can seamlessly adapt to unseen domains without additional retraining. Therefore, the generalization capability of CAFM is evaluated on unseen datasets to demonstrate its robustness.

4.3.1. Experimental design and result analysis

In this experiment, the CAFM model and two baseline models (CNN-LSTM and MLP) were trained on laboratory cycling data and then directly evaluated on four independent target datasets without fine-tuning or domain adaptation. The laboratory dataset was selected as the source domain because it was obtained under well-controlled conditions, providing high-quality, low-noise measurements that facilitate learning fundamental degradation patterns. The cross-dataset tasks included: Fig. 16(a) Lab-SNL (1C, 15 °C), Fig. 16(b) Lab-SNL (1C, 25 °C), Fig. 16(c) Lab-SNL (2C, 35 °C), and Fig. 16(d) Lab-CS2_35. These datasets were deliberately chosen to represent diverse degradation behaviors caused by variations in temperature, charge/discharge rate, and measurement platform. The first three SNL subsets allowed assessment under progressively increasing stress conditions, from mild degradation

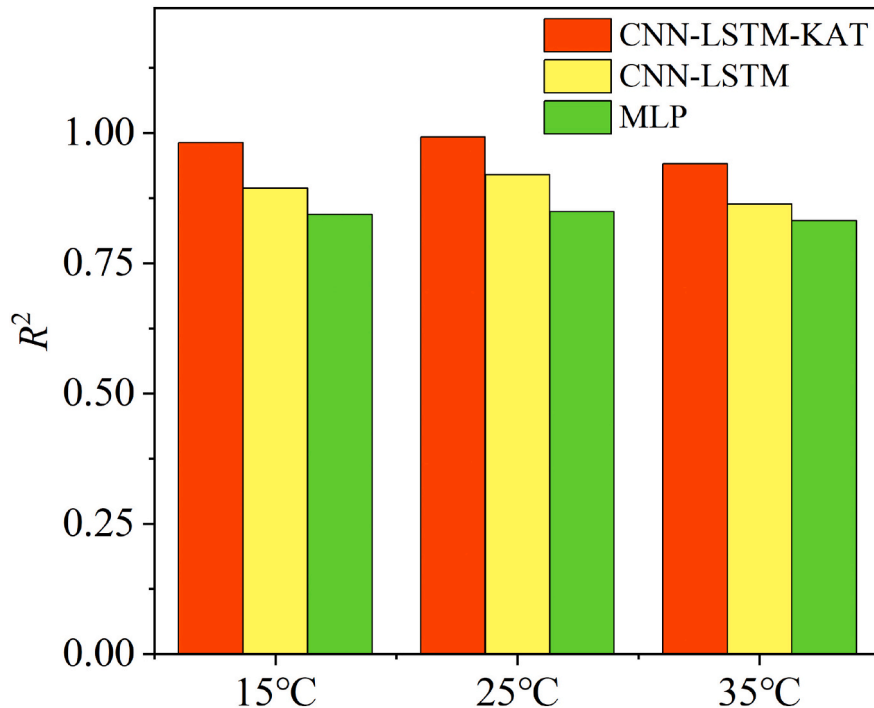


Fig. 13. R^2 for different models in the dataset at 1C.

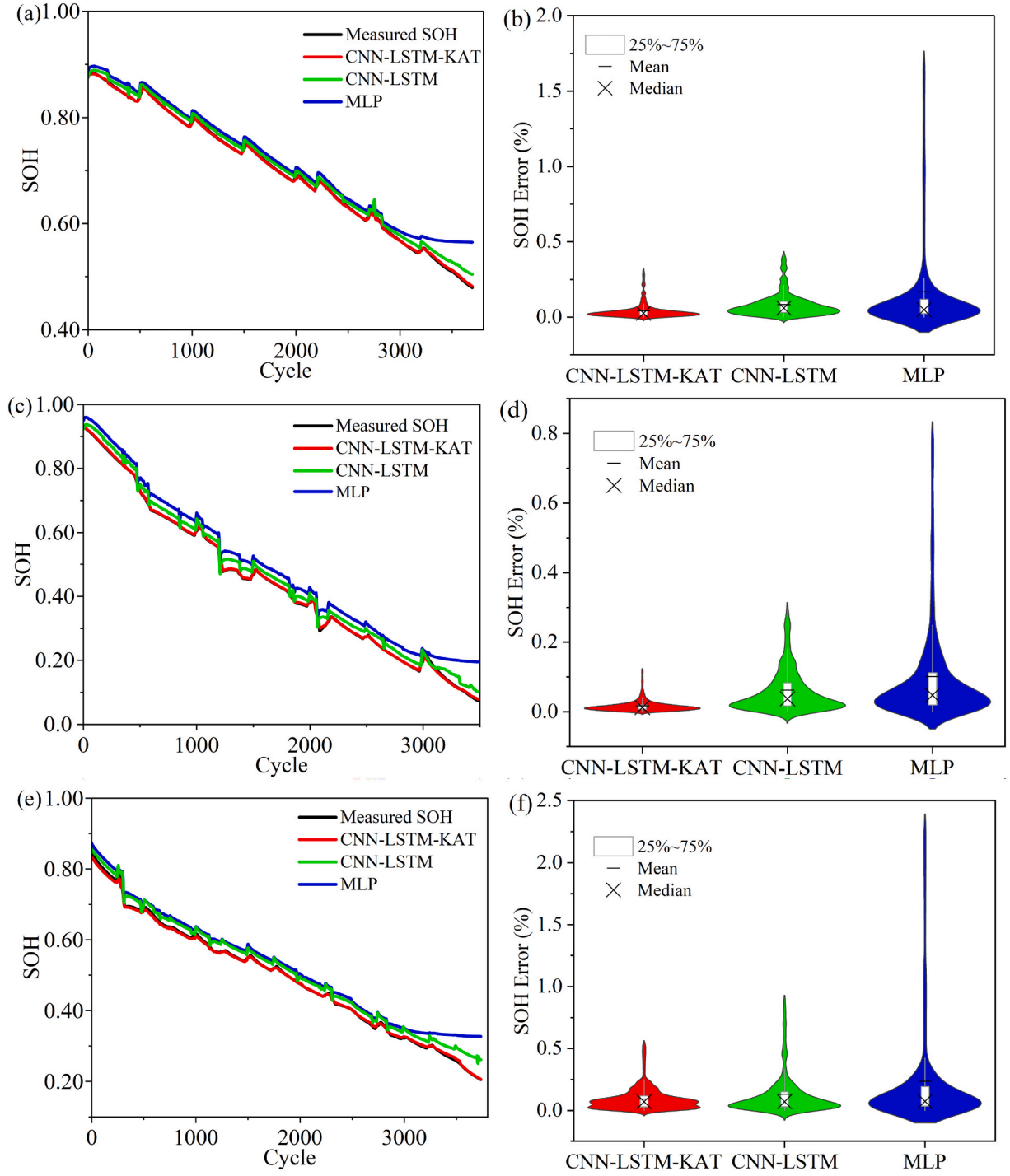


Fig. 14. SOH estimation results for 2C discharging rate at different temperatures. (a), (b) 15 °C. (c), (d) 25 °C. (e), (f) 35 °C.

Table 6

SOH estimation results for 2C discharging rate at different temperatures.

2C	Model	RMSE (%)	MAE (%)	R^2
15 °C	CAFM	0.56	0.41	0.9346
	CNN-LSTM	0.74	0.67	0.9057
	MLP	0.91	0.77	0.8834
25 °C	CAFM	0.31	0.38	0.9693
	CNN-LSTM	0.81	0.73	0.9438
	MLP	1.04	0.58	0.9184
35 °C	CAFM	0.73	0.71	0.8994
	CNN-LSTM	1.14	0.81	0.8125
	MLP	1.54	1.31	0.8491

at low temperature to accelerated aging at high temperature and high C-rate, testing model robustness across operating regimes. In contrast, CS2.35 originated from a distinct experimental platform with different instruments and protocols, enabling evaluation of true cross-platform generalization. For fair comparison, all models used identical HIs, featured normalization parameters, and evaluation metrics. Fig. 16 presents the SOH estimation results for these direct-transfer tasks, with corresponding quantitative metrics summarized in Table 7. This direct-transfer setup provides an objective evaluation of each model's intrinsic generalization capability, independent of retraining or domain adaptation.

In the Lab-SNL (1C, 15 °C) scenario, the CAFM model exhibited

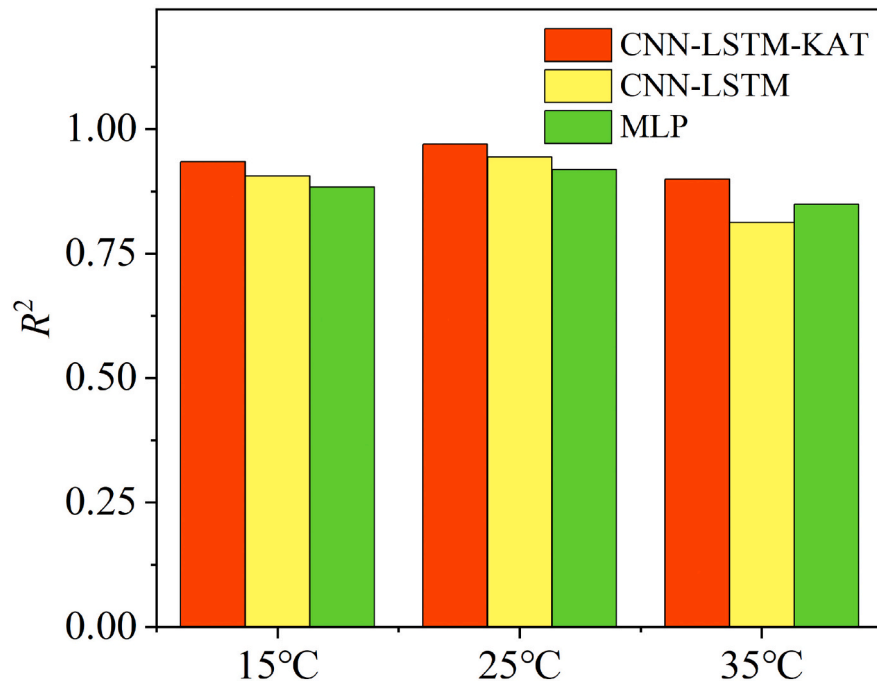


Fig. 15. R^2 for different models in the dataset at 2C.

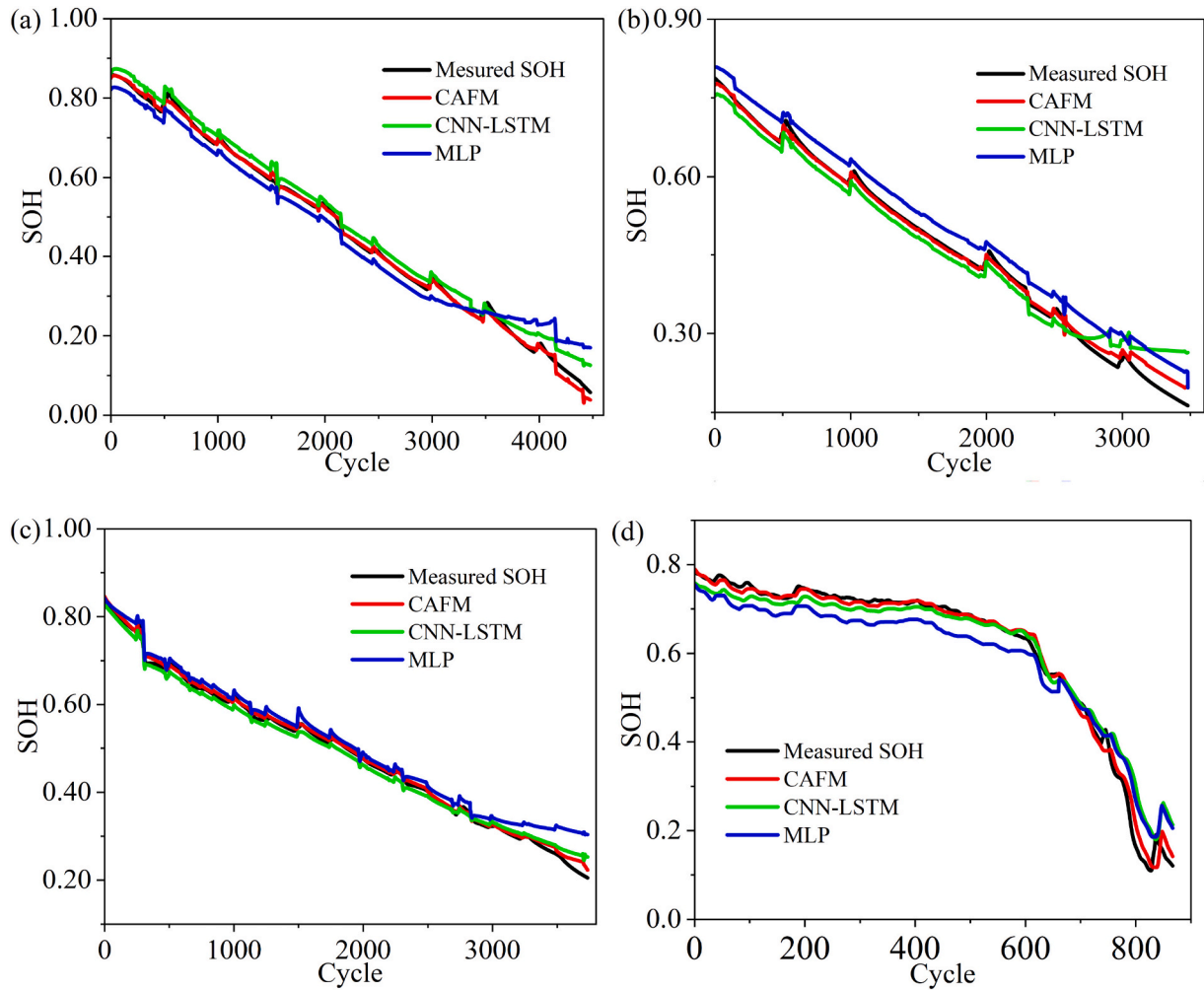


Fig. 16. SOH estimation results across datasets. (a) Lab-SNL (1C, 15 °C). (b) Lab-SNL (1C, 25 °C). (c) Lab-SNL (2C, 35 °C). (d) Lab-CS2_35.

Table 7
SOH estimation results across datasets.

Cross-dataset task	Model	RMSE (%)	MAE (%)	R^2
Lab-SNL (1C, 15 °C)	CAFM	0.77	0.62	0.9924
	CNN-LSTM	0.89	0.74	0.9720
	MLP	1.16	0.97	0.9610
Lab-SNL (1C, 25 °C)	CAFM	0.81	0.76	0.9911
	CNN-LSTM	0.76	0.68	0.9753
	MLP	1.25	1.04	0.9427
Lab-SNL (2C, 35 °C)	CAFM	0.60	0.51	0.9954
	CNN-LSTM	0.80	0.71	0.9842
	MLP	0.73	0.69	0.9794
Lab-CS2_35	CAFM	0.81	0.74	0.9910
	CNN-LSTM	0.94	0.71	0.9846
	MLP	0.98	0.82	0.9746

strong alignment with the measured SOH curve, accurately capturing both the overall degradation trend and local fluctuations. The CNN-LSTM model showed mild temporal delay in later cycles, while the MLP model underestimated degradation in the mid-life region. In the Lab-SNL (1C, 25 °C) direct-transfer, CAFM maintained stable prediction accuracy across all stages, effectively tracking small inflection points that other models tended to smooth out, indicating its robustness to moderate temperature perturbations. Under the more challenging Lab-SNL (2C, 35 °C) condition, where thermal effects and large current fluctuations accelerate degradation, CAFM demonstrated superior adaptability and maintained the closest fit to the measured SOH. In contrast, CNN-LSTM exhibited over-smoothing and lag in late-cycle predictions, while MLP's performance deteriorated significantly. When transferred to the CS2_35 dataset, which represents a cross-platform and cross-protocol scenario, CAFM still achieved high consistency with experimental data, maintaining minimal deviation across the entire life cycle.

Overall, CAFM consistently achieved the lowest RMSE and MAE and the highest R^2 among all compared models, confirming its superior domain robustness and generalization capability. The RMSE reduction relative to conventional baselines ranges from approximately 13 % to 35 %, depending on the target dataset and reference model. The largest improvement is observed against MLP in the Lab-SNL (1C, 25 °C), with an RMSE decrease of about 35 %. A minor exception occurs in the same cross-dataset task, where CNN-LSTM achieves a slightly lower RMSE (about 6.6 % difference), though CAFM provides a smoother and more stable fit across the entire life cycle. Under the more challenging condition of high temperature and large current (Lab-SNL, 2C, 35 °C), CAFM exhibits a clear advantage, reducing RMSE by around 25 % compared with CNN-LSTM. In the cross-platform Lab-CALCE task, CAFM maintains superior accuracy, achieving 15–20 % lower RMSE than the baselines. These results confirm that the proposed correlation-aware fusion mechanism effectively captures direct-transferable degradation features while mitigating dataset-specific noise, ensuring robust and stable generalization across heterogeneous operating conditions.

The cross-dataset experiments validate that the proposed CAFM model achieves reliable SOH estimation performance across datasets with significant variations in temperature, C-rate, and data acquisition systems. Its hierarchical feature correlation mechanism effectively suppresses low-relevance noise and preserves direct-transferable degradation representations, thereby improving zero-shot generalization without any retraining. In contrast, conventional architectures such as CNN-LSTM and MLP demonstrate limited adaptability to distributional shifts. These results demonstrate that CAFM not only attains high accuracy in in-domain predictions but also maintains strong robustness under out-of-domain and cross-platform conditions, highlighting its potential for practical deployment in battery health management systems.

5. Conclusions

In order to investigate the latent states embedded in HIs that exhibit low correlation with the SOH during capacity degradation, and to assess the impact of such information on the generalization performance of SOH estimation models. This paper proposes a combined parallel model architecture of KAT and CNN-LSTM, where Kolmogorov-Arnold networks uplift the low-correlation HIs to the high-dimensional interaction space, and then the KAN outputs are analyzed by transformer for cross-sequence correlation. Then the transformer analyzes the implicit relational encoding of KAN outputs across sequences of correlations, and replaces the linear layer in the transformer with KANLinear to realize the nonlinear attention mechanism and domain knowledge fusion, to solve the problem of insufficient sensitivity of traditional attention mechanism to weakly correlated features. CNN-LSTM captures the local mutation of battery capacity by CNN, and then models the capacity degradation pattern by LSTM to enhance the characterization ability of model. The conclusions are summarized as follows.

- 1) The accuracy and generalizability of the proposed method are verified in three datasets. Firstly, the CALCE dataset is used to validate the effectiveness of the proposed method, which reduces the MAE by 40 % and RMSE by 32 % compared with the comparison model. The results based on the laboratory data show that the proposed method not only improves the accuracy of SOH estimation, but also proves that there is no overfitting phenomenon with different training set ratios.
- 2) Reducing RMSE and MAE by an average of 70 % and 71 % across diverse discharging rates and temperature settings, the proposed model clearly demonstrates superior generalization performance.
- 3) CNN-LSTM and MLP are adopted as the models for the ablation experiment, but the inputs to the ablation experimental model are the three HIs of the first gradient, while the PCC of the three HIs of the proposed method are at three gradients. The accuracy is still much improved compared to the other two models, which proves that CAFM can mine the implicit relationship between low-correlation features and SOH and has good generalization ability across different datasets.

However, there are still some shortcomings in this study that require further research. The proposed method captures the capacity mutation during battery degradation by introducing CNN-LSTM, whereas KAT has difficulty in capturing this information in features with low PCC in the experiment. In the generalizability study of the model, experiments with boundary conditions such as extreme temperature (<0 °C or >40 °C) and high charge-discharging rate (>3C) are not adequately covered by the SNL dataset, which may result in reduced model performance, and will be the primary focus of future investigations.

CRedit authorship contribution statement

Yubin Gong: Writing – original draft, Methodology, Investigation. **Xinshu Lu:** Investigation, Formal analysis, Data curation. **Zixiang Su:** Software, Data curation. **Zhen Huang:** Resources. **Yong Quan:** Conceptualization. **Zhiqiang Zhang:** Supervision. **Tiancheng Ouyang:** Writing – original draft, Supervision, Methodology, Investigation.

Author statement

We confirm that this article has not been published previously; is not under consideration elsewhere; has the full consent of all authors; and that, if accepted, will not be published elsewhere in the same form, in English or in any other language, without the written consent of the Publisher.

Funding

This work was supported by the Guangxi Science Fund for Distinguished Young Scholars [grant numbers 2023GXNSFFA026013]; the Scientific Research and Technology Development Program of Guangxi (Key project) [grant numbers ZG2503980045]; the Science and Technology Plan Project of Liuzhou [grant numbers 2024AA0201B001]; Innovation Project of Guangxi Graduate Education [grant numbers YCSW2025142].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] Y. Wang, J. Tian, Z. Sun, et al., A comprehensive review of battery modeling and state estimation approaches for advanced battery management systems, *Renew. Sust. Energ. Rev.* 131 (2020) 110015, <https://doi.org/10.1016/j.rser.2020.110015>.
- [2] T. Ouyang, C. Wang, S. Jin, et al., Fuzzy information granulation for capacity efficient prediction in lithium-ion battery, *Renew. Sust. Energ. Rev.* 211 (2025) 115241, <https://doi.org/10.1016/j.rser.2024.115241>.
- [3] J. Wu, X. Cui, J. Meng, et al., Data-driven transfer-stacking-based state of health estimation for lithium-ion batteries, *IEEE Trans. Ind. Electron.* 71 (1) (2023) 604–614, <https://doi.org/10.1109/TIE.2023.3247735>.
- [4] C. Zhang, L. Tu, Z. Yang, et al., A CMMOG-based lithium-battery SOH estimation method using multi-task learning framework, *J. Energy Storage* 107 (2025) 114884, <https://doi.org/10.1016/j.est.2024.114884>.
- [5] T. Ouyang, P. Xu, J. Chen, et al., Improved parameters identification and state of charge estimation for lithium-ion battery with real-time optimal forgetting factor, *Electrochim. Acta* 353 (2020) 136576, <https://doi.org/10.1016/j.electacta.2020.136576>.
- [6] C. Lin, X. Tuo, L. Wu, et al., Physics-informed machine learning for accurate SOH estimation of lithium-ion batteries considering various temperatures and operating conditions, *Energy* (2025) 134937, <https://doi.org/10.1016/j.energy.2025.134937>.
- [7] Y. Li, B. Xiong, D.M. Vilathgamuwa, et al., Constrained ensemble Kalman filter for distributed electrochemical state estimation of lithium-ion batteries, *IEEE Trans. Ind. Inform.* 17 (1) (2020) 240–250, <https://doi.org/10.1109/TII.2020.2974907>.
- [8] T. Ouyang, Y. Gong, J. Ye, et al., State co-estimation for lithium-ion batteries based on multi-innovations online identification, *Renew. Sust. Energ. Rev.* 210 (2025) 115204, <https://doi.org/10.1016/j.rser.2024.115204>.
- [9] R. Xiong, L. Li, Z. Li, et al., An electrochemical model based degradation state identification method of lithium-ion battery for all-climate electric vehicles application, *Appl. Energy* 219 (2018) 264–275, <https://doi.org/10.1016/j.apenergy.2018.03.053>.
- [10] L. Chen, X. Bao, A.M. Lopes, et al., State of health estimation of lithium-ion batteries based on equivalent circuit model and data-driven method, *J. Energy Storage* 73 (2023) 109195, <https://doi.org/10.1016/j.est.2023.109195>.
- [11] P. Xu, X. Hu, B. Liu, et al., Hierarchical estimation model of state-of-charge and state-of-health for power batteries considering current rate, *IEEE Trans. Ind. Inform.* 18 (9) (2021) 6150–6159, <https://doi.org/10.1109/TII.2021.3131725>.
- [12] Z. Bao, J. Nie, H. Lin, et al., A global-local context embedding learning based sequence-free framework for state of health estimation of lithium-ion battery, *Energy* 282 (2023) 128306, <https://doi.org/10.1016/j.energy.2023.128306>.
- [13] L. Li, Y. Li, W. Cui, et al., A novel health indicator for online health estimation of lithium-ion batteries using partial incremental capacity and dynamic voltage warping, *J. Power Sources* 545 (2022) 231961, <https://doi.org/10.1016/j.jpowsour.2022.231961>.
- [14] J. Wu, L. Fang, J. Meng, et al., Optimized multi-source fusion based state of health estimation for lithium-ion battery in fast charge applications, *IEEE Trans. Energy Convers.* 37 (2) (2021) 1489–1498, <https://doi.org/10.1109/TEC.2021.3137423>.
- [15] Y. Wang, K. Li, P. Peng, et al., Health diagnosis for lithium-ion battery by combining partial incremental capacity and deep belief network during insufficient discharge profile, *IEEE Trans. Ind. Electron.* 70 (11) (2022) 11242–11250, <https://doi.org/10.1109/TIE.2022.3224201>.
- [16] M. Savargaonkar, I. Oyewole, A. Chehade, et al., Uncorrelated sparse autoencoder with long short-term memory for state-of-charge estimations in lithium-ion battery cells, *IEEE Trans. Autom. Sci. Eng.* 21 (1) (2022) 15–26, <https://doi.org/10.1109/TASE.2022.3222759>.
- [17] J. Tian, R. Xiong, W. Shen, et al., Deep neural network battery charging curve prediction using 30 points collected in 10 min, *Joule* 5 (6) (2021) 1521–1534, <https://doi.org/10.1016/j.joule.2021.05.012>.
- [18] D. Li, P. Liu, Z. Zhang, et al., Battery thermal runaway fault prognosis in electric vehicles based on abnormal heat generation and deep learning algorithms, *IEEE Trans. Power Electron.* 37 (7) (2022) 8513–8852, <https://doi.org/10.1109/TPEL.2022.3150026>.
- [19] J. Feng, F. Cai, X. Zhan, et al., Multi-timescale inconsistency evaluation and data-driven state of health prediction for circulating water-cooled series battery pack, *Measurement* 242 (2025) 115982, <https://doi.org/10.1016/j.measurement.2024.115982>.
- [20] Y. Li, K. Li, X. Liu, et al., A hybrid machine learning framework for joint SOC and SOH estimation of lithium-ion batteries assisted with fiber sensor measurements, *Appl. Energy* 325 (2022) 119787, <https://doi.org/10.1016/j.apenergy.2022.119787>.
- [21] W. Wang, G. Yang, M. Li, et al., State-of-health estimation for lithium-ion batteries based on Bi-LSTM-AM and LLE feature extraction, *Front. Energy Res.* 11 (2023) 1205165, <https://doi.org/10.3389/fenrg.2023.1205165>.
- [22] T. Ouyang, Y. Su, C. Wang, et al., Combined meta-learning with cnn-lstm algorithms for state-of-health estimation of lithium-ion battery, *IEEE Trans. Power Electron.* 39 (8) (2024) 10106–10117, <https://doi.org/10.1109/TPEL.2024.3398010>.
- [23] P. Ren, S. Wang, X. Chen, et al., A novel multiple training-scale dynamic adaptive cuckoo search optimized long short-term memory neural network and multi-dimensional health indicators acquisition strategy for whole life cycle health evaluation of lithium-ion batteries, *Electrochim. Acta* 435 (2022) 141404, <https://doi.org/10.1016/j.electacta.2022.141404>.
- [24] S. Pepe, F. Ciucci, Long-range battery state-of-health and end-of-life prediction with neural networks and feature engineering, *Appl. Energy* 350 (2023) 121761, <https://doi.org/10.1016/j.apenergy.2023.121761>.
- [25] J. Xu, J. Qu, H. Xu, Capacity estimation of lithium-ion batteries with automatic feature extraction and graph-enhanced LSTM, *J. Energy Storage* 85 (2024) 111131, <https://doi.org/10.1016/j.est.2024.111131>.
- [26] J. Tian, R. Xiong, W. Shen, A review on state of health estimation for lithium ion batteries in photovoltaic systems, *ETransportation* 2 (2019) 100028, <https://doi.org/10.1016/j.etrans.2019.100028>.
- [27] Y. Ma, C. Shan, J. Gao, et al., A novel method for state of health estimation of lithium-ion batteries based on improved LSTM and health indicators extraction, *Energy* 251 (2022) 123973, <https://doi.org/10.1016/j.energy.2022.123973>.
- [28] Y. Tang, K. Yang, H. Zheng, et al., Early prediction of lithium-ion battery lifetime via a hybrid deep learning model, *Measurement* 199 (2022) 111530, <https://doi.org/10.1016/j.measurement.2022.111530>.
- [29] W. Zhao, W. Ding, S. Zhang, et al., Enhancing lithium-ion battery lifespan early prediction using a multi-branch vision transformer model, *Energy* 302 (2024) 131816, <https://doi.org/10.1016/j.energy.2024.131816>.
- [30] Z. Zhang, Y. Wang, X. Ruan, et al., Lithium-ion batteries lifetime early prediction using domain adversarial learning, *Renew. Sust. Energ. Rev.* 208 (2025) 115035, <https://doi.org/10.1016/j.rser.2024.115035>.
- [31] M. Miao, C. Xiao, J. Yu, Multi-source domain meta-learning network for battery state-of-health estimation under multi-target working conditions, *IEEE Trans. Power Electron.* 40 (7) (2025) 9786–9799, <https://doi.org/10.1109/TPEL.2025.3538469>.
- [32] C. Lin, J. Xu, X. Mei, Improving state-of-health estimation for lithium-ion batteries via unlabeled charging data, *Energy Storage Mater.* 54 (2023) 85–97, <https://doi.org/10.1016/j.ensm.2022.10.030>.
- [33] Z. Liu, Y. Wang, S. Vaidya, et al., Kan: Kolmogorov-Arnold Networks. arXiv preprint arXiv, 2024, <https://doi.org/10.48550/arXiv.2404.19756>, 2404.19756.
- [34] Z. Zhang, R. Zhang, X. Liu, et al., Advanced state-of-health estimation for lithium-ion batteries using multi-feature fusion and KAN-LSTM hybrid model, *Batteries* 10 (12) (2024) 433, <https://doi.org/10.3390/batteries10120433>.
- [35] J. Peng, X. Zhao, J. Ma, et al., State of health estimation of lithium battery via incremental capacity analysis and internal resistance identification based on Kolmogorov-Arnold networks, *Batteries* 10 (9) (2024), <https://doi.org/10.3390/batteries10090315>.
- [36] C. Chen, Y. Wu, J. Shi, et al., A parallel weighted ADTC-transformer framework with FUnet fusion and KAN for improved lithium-ion battery SOH prediction, *Control. Eng. Pract.* 159 (2025) 106302, <https://doi.org/10.1016/j.conengprac.2025.106302>.
- [37] T. Ouyang, C. Wang, P. Xu, et al., Prognostics and health management of lithium-ion batteries based on modeling techniques and Bayesian approaches: a review, *Sustain. Energy Technol. Assess.* 55 (2023) 102915, <https://doi.org/10.1016/j.seta.2022.102915>.
- [38] P. Xu, C. Wang, J. Ye, et al., State-of-charge estimation and health prognosis for lithium-ion batteries based on temperature-compensated Bi-LSTM network and integrated attention mechanism, *IEEE Trans. Ind. Electron.* 71 (6) (2023) 5586–5596, <https://doi.org/10.1109/TIE.2023.3292865>.
- [39] T. Zhu, W. Wang, M. Yu, A novel hybrid scheme for remaining useful life prognostic based on secondary decomposition, BiGRU and error correction, *Energy* 276 (2023) 127565, <https://doi.org/10.1016/j.energy.2023.127565>.
- [40] Z. Zhu, Q. Yang, X. Liu, et al., Attention-based CNN-BiLSTM for SOH and RUL estimation of lithium-ion batteries, *J. Algorithms Comput. Technol.* 16 (2022) 17483026221130598, <https://doi.org/10.1177/17483026221130598>.
- [41] Q. Liu, H. Sun, R. Chen, et al., Research on the state of health estimation of lithium-ion battery based on bidirectional long short-term memory neural network and particle swarm optimization algorithm, *J. Energy Storage* 138 (2025) 118756, <https://doi.org/10.1016/j.est.2025.118756>.